SC- and PC-NEVPT2 and QD-NEVPT2 in free-breakfast

Energies, analytic nuclear gradients and Hessians, and nonadiabatic couplings, density-fitted, on the GPU

free-breakfast implementation notes

8 November 2026

Abstract

This document describes the `src/nevpt2` layer of free-breakfast: the strongly contracted (SC) and partially contracted (PC) n-electron valence state perturbation theory (NEVPT2) energies of a CASSCF reference, their quasidegenerate (QD) effective Hamiltonians over a state-averaged CASSCF, the analytic nuclear gradient of every one of them, the analytic nuclear Hessian of single-state SC and PC and of an SC-QD state, the QD interstate coupling and derivative (nonadiabatic) coupling, and what a QD minimum-energy conical intersection search needs. Everything is density fitted and runs on the device. The organising idea is that every correlation quantity is a function of only three objects, the MO one-electron matrix h, the fitted three-index tensor B and the CI vectors, so one reverse pass through the energy gives its adjoints $(\bar{h}, \Lambda, \bar{y})$, and from there each derivative is the CASSCF machinery the tree already has: one Z-vector solve over the CI-relaxed Hessian and one or two passes of the density-fitted derivative integrals. The Hessian is the same idea one order up: the energy's tangent along each coordinate's first-order response, forward over the reverse pass, against $3N$ CP-CASSCF responses and the gradient's own multipliers; a QD state's adds one closed-form eigenvector term built from the interstate couplings. No derivative is written by hand beyond a few scalar finishes, and no finite difference appears in any production path except those that are exact: the Hessian-vector products' bars at fixed multipliers are polynomials of degree at most five along a line, and three symmetric differences recover their derivative with no truncation error.

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1 Scope and layering

What is implemented.

- SC- and PC-NEVPT2 class energies of one root [1, 2, 3], over a Dyall zeroth-order Hamiltonian [5].
- SC- and PC-QD-NEVPT2 [4]: the effective Hamiltonian of n roots of a state-averaged CASSCF, in the "SA-NEVPT2" scheme of Park [10] (one set of orbitals, core and virtuals canonical over the state-averaged Fock), symmetrised and diagonalised.
- Analytic nuclear gradients of single-state SC and PC-NEVPT2 on a single-state DF-CASSCF [8, 9], and of any SC- or PC-QD-NEVPT2 state on a DF-SA-CASSCF at any weights [10].
- Analytic nuclear Hessians of single-state SC- and PC-NEVPT2 on a single-state DF-CASSCF, all-electron, in Cartesian or caller-given coordinates (§7), with the refusals that keep a non-smooth surface from being reported as a Hessian. Unpublished for any internally contracted MRPT2 as far as we know.
- The analytic nuclear Hessian of an SC-QD-NEVPT2 state on a DF-SA-CASSCF at any weights (§11), with every model state's refusals and a report of every small denominator (§12); and the semi-numerical QD Hessian in the pipeline (§13). PC-QD's analytic Hessian (#966), the pair-mean Hessian at a crossing (#967) and the analytic `[hessian]` over a QD chain (#968) are not yet in the tree.
- The QD interstate coupling $\mathbf{h}_{PQ}$, and the full derivative coupling $\mathbf{d}_{PQ} = \langle \Psi_P | \nabla \Psi_Q \rangle$, including the PT2 term [10, 11].
- For conical-intersection searches: $\mathbf{g}_P$, $\mathbf{g}_Q$ and $\mathbf{h}_{PQ}$ at one set of H^{eff} eigenvectors with no refusal of a degenerate pair, and bounds on how far the pair's gap moves when the SA-CASSCF is not fully converged; for PC, the mean gradient alone inside a degenerate CASSCF pair, where the branching plane has no limit.

Layering. `src/nevpt2` sits over `core`, `integrals` (the density-fitted MO transform), `ci` (the strings, sigma vectors and RDMs) and, for derivatives, `response` and `mcscf` (the CASSCF Fock, CI-relaxed Hessian and Z-vector in `response`; the DF gradient, derivative integrals and CASSCF couplings in `mcscf`). It never imports `pipeline` or `hessian`. The nuclear Hessian is assembled one layer up, in `hessian` (`df_sc_nevpt2`, `df_pc_nevpt2`, `df_qd_sc_nevpt2`), from this layer's tangents and Hessian-vector products and the CASSCF Hessian's parts. The pipeline's `[nevpt2]`, `[nevpt2_gradient]`, `[nevpt2_nac]`, `[optimize]` and `[hessian]` steps wrap both. Its six sub-modules are:

module	role
`nevpt2.density_fitting`	MO blocks of B, the active $(tu\|vw)$, the ten external blocks slab by slab, and their adjoints onto Λ (§4.1)
`nevpt2.rdm`	dm3 and the integral-contracted 4-RDMs f3ca, f3ac, of one vector or between two, and their adjoints (§4.2)
`nevpt2.energy`	the eight SC and PC class energies, the SC- and PC-QD effective Hamiltonians, the amplitude hand-over, the reverse passes of all four, and their forward tangents and Hessian-vector products, single-state and QD (§§4–5, §7.3, §11.3)
`nevpt2.gradient`	the energy as a function of (h, B, C, c), z^c, the Lagrangians, Z-vectors and nuclear terms; single-state and QD, SC and PC (§§6–8); the builders' forward tangent and the Hessian-vector product in (h, B, c), and their QD forms over a state average (§7, §11)
`nevpt2.nac`	the PT2 transition density and the QD derivative coupling (§9)
`nevpt2.intersection`	a QD pair's branching space and gap sensitivities (§10)

Provenance. The energy kernels are a port of the nevpt2-gpu-demo, itself a transcription of PySCF's `mrpt.nevpt2` [16]: every einsum keeps the reference's subscripts, so the two read side by side. The PC energy is validated against block2's `WickICNEVPT2` [17]. The derivative layers are this tree's own. A related device-resident approach, DMRG-QD-SC-NEVPT2 gradients and couplings as one reverse-mode transpose of a contraction graph, is Ref. [18].

2 Notation

Orbitals. C ($n_{\mathrm{ao}} \times n_{\mathrm{mo}}$) maps AOs μ, ν to MOs, ordered core, active, virtual. Core (inactive) labels are i, j, k, l, active t, u, v, w (and p, q, r, s inside active-only tensors), virtual a, b (and r, s as the external labels of a class), general x, y, z. Counts are n_c, n_a, n_v, n_{mo}. A rotation is $C \to Ce^{\kappa}$, κ real antisymmetric.

Integrals. $h = C^{\mathsf{T}} h^{\mathrm{AO}} C$. With three-centre integrals $(\mu\nu|P)$ and the Coulomb metric $(P|Q) = LL^{\mathsf{T}}$,

$$B^L_{xy} = \sum_P (L^{-1})_{LP} \left(C^{\mathsf{T}}(\cdot\,\cdot|P)\,C\right)_{xy}, \qquad (xy|zw) = \sum_L B^L_{xy} B^L_{zw}, \tag{1}$$

the density-fitting approximation [15]. Every two-electron quantity below is built from B through such products.

Densities. Spin-summed E_{xy}; active RDMs $\Gamma^{(1)}_{tu} = \langle E_{tu}\rangle$, dm2 $= \langle E_{tu}E_{vw}\rangle$, dm3 $= \langle E_{tu}E_{vw}E_{xy}\rangle$, in PySCF's unreordered layout. Between a bra $\langle M|$ and a ket $|N\rangle$, $\gamma^{(MN)}_{tu} = \langle M|E_{tu}|N\rangle$ and so on, unsymmetrised.

Fock matrices. The inactive Fock and the generalised Fock are

$$F^I_{xy} = h^{\mathrm{eff}}_{xy} = h_{xy} + \sum_j \big[2(xy|jj) - (xj|jy)\big], \qquad f_{xy} = F^I_{xy} + \sum_{tu} \gamma_{tu}\big[(xy|tu) - \tfrac{1}{2}(xt|uy)\big], \tag{2}$$

and the orbital energies are $\varepsilon_x = f_{xx}$ for x core or virtual. On a state average $\gamma \to \gamma^{\mathrm{SA}} = \sum_r w_r \Gamma^{(1)}_r$.

3 NEVPT2 and QD-NEVPT2

3.1 The zeroth-order Hamiltonian and the classes

NEVPT2 partitions $H = H^{(0)} + V$ with Dyall's Hamiltonian [5]

$$H^D = \sum_i \varepsilon_i E_{ii} + \sum_a \varepsilon_a E_{aa} + \sum_{tu} h^{\text{eff}}_{tu} E_{tu} + \tfrac{1}{2} \sum_{tuvw} (tu|vw)\big(E_{tu}E_{vw} - \delta_{uv}E_{tw}\big) + C, \tag{3}$$

and splits the perturbation into eight classes $V_l^{(k)}$ by how many electrons the active space gains (k) and from which external orbitals (l) [1]. Their PySCF names are used throughout:

name	space	external tuple	Δ (orbital-energy part of the denominator)
Sr	$S_r^{(-1)}$	a	ε_a
Si	$S_i^{(+1)}$	i	$-\varepsilon_i$
Sijrs	$S_{ijab}^{(0)}$	(i,j,a,b)	$\varepsilon_a + \varepsilon_b - \varepsilon_i - \varepsilon_j$
Sijr	$S_{ija}^{(+1)}$	(i,j,a)	$\varepsilon_a - \varepsilon_i - \varepsilon_j$
Srsi	$S_{abi}^{(-1)}$	(a,b,i)	$\varepsilon_a + \varepsilon_b - \varepsilon_i$
Srs	$S_{ab}^{(-2)}$	(a,b)	$\varepsilon_a + \varepsilon_b$
Sij	$S_{ij}^{(+2)}$	(i,j)	$-\varepsilon_i - \varepsilon_j$
Sir	$S_{ia}^{(0)}$	(i,a)	$\varepsilon_a - \varepsilon_i$

Every class but Sijrs has active indices; Sijrs reads no RDM.

3.2 The SC energy

In SC-NEVPT2 each external tuple t has one perturber, $P_tV|\Psi_0\rangle$. Its norm $N_t = \langle\Psi_0|V^\dagger P_t V|\Psi_0\rangle$ and its Dyall element H_t (the active part of $\langle P_tV\Psi_0|H^D - E_0|P_tV\Psi_0\rangle$) are quadratic forms of a slab row of integrals over RDM-built matrices, and

$$E^{\text{SC}} = \sum_k \sum_{t\in k} E_t, \qquad E_t = -\frac{N_t}{\Delta_t + H_t/N_t}, \tag{4}$$

over the perturbers with $|N_t| > 10^{-14}$ (PySCF's `_norm_to_energy` guard; Sijrs and the one-function classes below have none). The matrices are the metrics (dm3, dm2, dm1 and their hole forms) and the Dyall blocks `a16`, `a22`, ... of PySCF, which for Sr and Si read the 4-RDM only through f3ca and f3ac (§4.2).

One-function classes (#949). Srs at two active electrons and Sij at two active holes span one function per perturber: removing both electrons leaves the active vacuum (filling both holes, the closed shell), so $N_t \propto A_t^2$ for one amplitude $A_t = \sum_{pq} W_{pq}\langle\text{vac}|a_q a_p|\Psi\rangle$ and $H_t = \eta N_t$, η that function's Dyall energy, one number per state and class. Every CAS(2,2) has both. PySCF's guard drops such a perturber wherever A_t, a smooth signed function of the geometry, passes through zero, yet its QD couplings $T^t_{IJ} \propto A^I_t A^J_t$ are linear in A_t there: that was a 3–6 $\times 10^{-8}$ E_h/bohr SC-QD gradient miss at ethylene's Py (#894). So a one-function class (told by tr dm1: Srs at 2, Sij at $2n_a - 2$) keeps every perturber, with $c_t = -1/(\Delta_t + \eta)$ and

$$\eta = \frac{\langle X, Y\rangle}{\langle Y, Y\rangle}, \tag{5}$$

X the class's Dyall block (a7, a9) and Y its metric (rm2, hdm2) in the pairing the H and norm einsums contract. On a state's own RDMs $X = \eta Y$ to 10^{-15}, so η is every perturber's H_t/N_t read with no division by a small N_t. Off that structure (a raw direction in a test) η is the least-squares ratio, one smooth function that every pass differentiates exactly. PC has no guard and is unchanged.

3.3 The PC energy

PC-NEVPT2 keeps the span of each class's internal functions [3]. For a class with metric S, Dyall matrix K, one-body metric S_1 and a coupling factor c, a tuple's slab row x_t and its one-body row h_t give

$$E_t = -c\, b_t^{\mathsf{T}} (K + \Delta_t S)^{+} b_t, \qquad b_t = S x_t + S_1 h_t, \tag{6}$$

$(\cdot)^+$ the inverse on the span of S that the metric cut keeps. S, K and S_1 are the same matrices SC builds (SC contracts each against one row; PC solves with them), so PC is "SC without the final contraction":

class	basis, d	S	K	S_1	c
Sr	(p,q,r), n_a^3	dm3$[r,p,q,b,a,c]$	a16	dm2$[r,p,q,a]$	1
Si	(p,q,r), n_a^3	dm3_h$[r,p,q,b,a,c]$	a22	dm2_h$[r,p,q,a]$	1
Sijr	p, n_a	hdm1	a3	—	$2\,\mathrm{direct}-\mathrm{exchange}$
Srsi	p, n_a	dm1	k27	—	$2\,\mathrm{direct}-\mathrm{exchange}$
Srs	(p,q), n_a^2	rm2$[p,q,b,a]$	a7	—	$\frac{1}{2}$
Sij	(p,q), n_a^2	hdm2	a9	—	$\frac{1}{2}$
Sir	$(p,q)\oplus(p,q)$, $2n_a^2$	spin-coupling blocks	a12, a13	$[2\,\mathrm{dm1};\ -\mathrm{dm1}]$	1
Sijrs	none	—	—	—	SC's

3.4 QD-NEVPT2

Over n roots $|M\rangle$ of a state-averaged CASSCF sharing one set of orbitals, with ε from f^{SA} for every root (Zaitsevskii–Malrieu multipartitioning [6], Park's SA-NEVPT2 scheme [10]), each root has its own perturber space and Dyall Hamiltonian, and the couplings are $C_{MN} = \langle M|V|\Psi_N^{(1)}\rangle$. With $\Psi_N^{(1)}$ the ket's first-order function:

$$\text{SC:}\quad C_{MN} = -\sum_t \frac{\bar{T}^t_{MN}}{\Delta_t + H_t^N/N_t^N}, \qquad \bar{T}^t_{MN} = \tfrac{1}{2}(T^t_{MN} + T^t_{NM}), \tag{7}$$

$$\text{PC:}\quad C_{MN} = -c\sum_t b_t^{M\leftarrow N\,\mathsf{T}} (K_N + \Delta_t S_N)^{+} b_t^N, \qquad b_t^{M\leftarrow N} = S^{(N,M)} x_t + S_1^{(N,M)} h_t. \tag{8}$$

T^t_{MN} is the class's norm einsum over pair (M,N)'s transition RDMs; the terms of it that read no RDM carry $\langle M|N\rangle = 0$ and are dropped. The mean over (M,N) and (N,M) in SC undoes PySCF's factor 2 on the Hermitian cross terms of Sr, Si and Sir, which holds only for a state's own RDMs. $S^{(N,M)}$, $S_1^{(N,M)}$ are the PC metrics over pair (N,M)'s transition RDMs (bra N, ket M); S_1 must be that orientation. PC has no mean, so its C is not symmetric. Sijrs is diagonal. The effective Hamiltonian is symmetrised as in MS-CASPT2 [10]:

$$H^{\mathrm{eff}}_{MN} = E_M \delta_{MN} + \tfrac{1}{2}(C_{MN} + C_{NM}), \qquad H^{\mathrm{eff}} R_P = E_P R_P, \quad R^{\mathsf{T}} R = 1, \tag{9}$$

each column of R signed so its largest-magnitude component is positive.

4 The energy on the device

4.1 Density-fitted integrals, slab by slab

`mo_blocks` forms the five MO blocks B_{aa}, B_{ca}, B_{va}, B_{cv} and B_{vc} (B_{cv} with its pair swapped, so every batched index is the first of its pair). No $(Q|vv)$ block exists: no class has two virtuals in one charge distribution. The ten external blocks are built a slab at a time along their batch axis, so no whole external block is ever on the device:

$$\mathcal{C}_{xyzw} = (xy|zw) = \sum_Q X_{Q,xy}\, Y_{Q,zw}, \qquad \text{slab} = \text{permute}(\mathcal{C}), \tag{10}$$

one strided-batched GEMM over y ($\mathcal{C}_y = X_y^{\mathsf{T}} Y$) and one permute kernel per slab; Sijrs needs no permute.

block	physicist slab	X	Y	axes	batch
Sr	$[r,a,b,c] = (rb\|ac)$	B_{va}	B_{aa}	0213	0
Si	$[t,u,i,v] = (it\|uv)$	B_{ca}	B_{aa}	1203	2
Sijrs	$[i,a,j,b] = (ia\|jb)$	B_{cv}	B_{cv}	0123	0
Sijr	$[r,p,j,i] = (rj\|ip)$	B_{vc}	B_{ca}	0312	0
Srsi	$[x,y,i,t] = (xi\|yt)$	B_{vc}	B_{va}	0213	0
SrsiT	$[y,x,i,t] = (xt\|yi)$	B_{va}	B_{vc}	2031	1
Srs	$[r,s,t,u] = (rt\|su)$	B_{va}	B_{va}	0213	0
Sij	$[t,u,i,j] = (it\|ju)$	B_{ca}	B_{ca}	1302	2
Sir1	$[r,t,i,u] = (ri\|tu)$	B_{vc}	B_{aa}	0213	0
Sir2	$[r,u,t,i] = (rt\|iu)$	B_{va}	B_{ca}	0312	0

Adjoints. With $\bar{\mathcal{C}}$ the slab bar permuted back, $\bar{X}_y \mathrel{+}= Y\bar{\mathcal{C}}_y^{\mathsf{T}}$ and $\bar{Y}_w \mathrel{+}= X_b\bar{\mathcal{C}}_w$, two strided-batched GEMMs written straight into $\Lambda = \partial E/\partial B$ ($n_{\text{aux}} \times n_{\text{mo}}^2$, B's own layout) at each block's corner, so no block bar is ever formed. A slab's bar is consumed by its call and freed before the next slab is built.

4.2 RDMs: dm3 and the contracted 4-RDMs

The 4-RDM never exists. Sr and Si read it only contracted with $(tu|vw)$:

$$\text{f3ca}_{pqrstu} = \sum_{ijk} (ki|uj)\, \langle E_{pq}E_{rs}E_{tj}E_{ki}\rangle, \qquad \text{f3ac}_{pqrstu} = \sum_{ijk} (ik|jt)\, \langle E_{pq}E_{rs}E_{ju}E_{ik}\rangle, \tag{11}$$

(here i,j,k active). `nevpt2::rdms` builds dm3, f3ca and f3ac (n_a^6 each), tiled over determinants K against a byte budget. Per tile, one thread per determinant gathers

$$R_{tu,K} = \langle K|E_{tu}|c\rangle, \qquad L2_{pqrs,K} = \langle c|E_{pq}E_{rs}|K\rangle \tag{12}$$

from the strings' own replacement lists (Knowles–Handy [14]), then dm3 $\mathrel{+}= L2\, R^{\mathsf{T}}$ is one GEMM; per order the consume $W = (tu|vw)\cdot L2$ is n_a strided-batched GEMMs and $f3 \mathrel{+}= L2\, W^{\mathsf{T}}$ one GEMM. After the last tile a contraction of $(tu|vw)$ with dm3 (fdm2) and PySCF's wedge reconstruction (the commutator $E_{pq}E_{rs} = E_{rs}E_{pq} + \delta_{qr}E_{ps} - \delta_{ps}E_{rq}$) run once per order. Nothing is atomic, so the result does not depend on the run. The cost is $O(N_{\text{det}} n_a^8)$ in GEMMs, the same order as forming the 4-RDM, with $O(N_{\text{det}} n_a^4)$ memory per tile.

Between two vectors. The bra/ket build gathers $L2$ over the bra and R over the ket, so $L2\,R^{\mathsf{T}} = \langle M|EEE|N\rangle$; dm1 and dm2 come from one GEMV each per tile. The consume reads the ket's own $L2$ (a second gather), and the ca order's fdm2 reads the transition element that hermiticity let the one-vector build skip. With `bra == ket` it is the one-vector build operation for operation. QD reads f3 on the diagonal only, so the pair loop passes null f3 pointers and only dm1–dm3 are built, about a third of the GEMM work.

Adjoints. `rdms_adjoint` runs the build backwards over the same tiles: the wedge and fdm2 first, then per tile each digest acc $+= L2\,B^{\mathsf{T}}$ as two GEMMs ($\bar{L2} += \overline{\mathrm{acc}}\,B$, $\bar{B} = \overline{\mathrm{acc}}^{\mathsf{T}} L2$), each consume as two loops of batched GEMMs, and one gather onto the CI bar over each determinant's replacement lists. dm1 and dm2 have no bars of their own: since $\sum_t E_{tt} = N$ on the ket,

$$\mathrm{dm2}_{pqrs} = \tfrac{1}{N}\sum_t \mathrm{dm3}_{pqrstt}, \qquad \mathrm{dm1}_{pq} = \tfrac{1}{N^2}\sum_{rt} \mathrm{dm3}_{pqrrtt}, \tag{13}$$

so their bars fold into $\overline{\mathrm{dm3}}$ along those traces, and one adjoint carries every RDM. Neither a sigma vector nor any symmetry of a bar is assumed.

4.3 The SC and PC class walks

Each class builds its RDM-only intermediates once, then walks its external index in batches: the slab is built, contracted by the class's einsums (`recipes::einsum`, with the reference's subscripts unchanged) into per-perturber N_t and H_t, read back and finished on the host, Eq. (4). Srsi pairs (r, s) with (s, r) across slabs, so it fills its whole (v, v, c) tables first.

PC solves each class once, since S, K and S_1 do not depend on the tuple (Algorithm 1). Each slab is then one einsum into the eigenbasis.

Algorithm 1 A PC class (`solve_class`, then the walk)

Input: S, K, S_1 ($d \times d$, $d \times n_1$) of one state; the class's slabs

1: $S \leftarrow \mathrm{sym}\,S$, $K \leftarrow \mathrm{sym}\,K$ ▷ K is symmetric only to the CI's convergence
2: $S = V\sigma V^{\mathsf{T}}$; refuse if $\sigma_{\max} \le 0$; keep $\sigma_k > \max(10^{-13}, 100\,\varepsilon_{\mathrm{mach}}\sqrt{d}, f)\,\sigma_{\max}$ ▷ no gap test (#853); f the floor
3: $X \leftarrow V_{\mathrm{kept}}\sigma_{\mathrm{kept}}^{-1/2}$; $K' = X^{\mathsf{T}}KX = W\lambda W^{\mathsf{T}}$; $U \leftarrow XW$
4: $\mathcal{T} \leftarrow SU$; $t_1 \leftarrow S_1^{\mathsf{T}}U$
5: **for** each slab, each tuple t **do**
6: $\quad y_t \leftarrow \mathcal{T}^{\mathsf{T}}x_t + t_1^{\mathsf{T}}h_t$ ▷ one einsum per slab
7: $\quad$ refuse if $\min_k(\lambda_k + \Delta_t) \le 0$
8: $\quad E_t \leftarrow -c\sum_k y_{tk}^2/(\lambda_k + \Delta_t)$

S is singular by construction (Sr and Si drop 44 of 64 modes at CAS(4,4)). The cut sits at least 13 times above every mode round-off or the RDMs decide, measured; above it the real modes may run through the cut with no gap, and carry no energy there (PSB3, HF with its F 1s active: the energy moves by under $3 \times 10^{-9}\,E_h$ with the cut anywhere in $10^{-15} \ldots 10^{-12}$; `src/nevpt2/energy/README.md`, #853), so there is no gap test. A refused class (no positive σ, or a non-positive denominator) fails the call (naming the class, and for QD the state); it is never summed in as zero.

The conditioning floor (#917, #923, #1020). f is `EnergyInputs::pc_floor`, relative to each metric's $\sigma_{\max}$, and each class reports `cut` (the largest dropped σ) and `kept` (the smallest kept), over $\sigma_{\max}$. The library default is $f = 0$, the round-off cut alone, so every energy, gradient and golden is bitwise what it was before the floor existed. The floor matters at second order, not first. At PSB3, DF-CASSCF(6,6)/cc-pVDZ, real modes sit at about 5×10^{-13} of $\sigma_{\max}$. Along the Hessian's physical directions the first-order bars repeat to 10^{-10} of their largest component, but a central-difference ladder of the PC adjoint never converges: near-cut kept modes swing λ through $-\Delta$ as $1/\sigma$. At $f = 10^{-10}$ the ladder is clean at FC and at the 90° twist. The energy moves by 1.2×10^{-10} (FC) and $5.8 \times 10^{-9}\,E_h$ (twist), and about 1.7×10^{-7} at HF CAS(10,7). A library-wide 10^{-10} would move the PC-QD goldens and NACs past their bars, so it is applied only where needed: the pipeline's PC `[hessian]` runs its whole chain at `pc_floor` $= 10^{-10}$ by default. The QD entry points take a trailing `pc_floor = 0` and set it on every state's inputs; nothing in the pipeline raises it there. One field read by the one `solve_class` means the energy, its reverse pass and its tangent always differentiate the same function.

4.4 The QD walks and the amplitude hand-over

`sc_effective_hamiltonian` and `pc_effective_hamiltonian` take $n \times n$ pair inputs, entry (M, N) carrying the shared integrals and pair (M, N)'s transition dm1–dm3 (f3 on the diagonal only). Each class walks its slabs once for every pair. For SC, per slab, one table per given pair accumulates T^t_{MN} and one per given state N^N_t, H^N_t; the Dyall blocks are built from each state's own RDMs only. For PC, each ket N's $(S_N, K_N, S_{1,N})$ is solved once, the transitions' $\mathcal{T}_{M\leftarrow N} = S^{(N,M)\mathsf{T}} U_N$ and $t_{1,M\leftarrow N} = S_1^{(N,M)\mathsf{T}} U_N$ are built once per pair, and

$$C^t_{MN} = -c \sum_k \frac{y_{M\leftarrow N,tk}\; y_{N,tk}}{\lambda_{N,k} + \Delta_t}, \qquad y_{M\leftarrow N,t} = \mathcal{T}^{\mathsf{T}}_{M\leftarrow N} x_t + t^{\mathsf{T}}_{1,M\leftarrow N} h_t, \tag{14}$$

with Sijr and Srsi's "2 direct − exchange" as the bilinear $\sum_t (2 y_{M\leftarrow N,t} y_{N,t} - y_{M\leftarrow N,t} y_{N,t'})/(\lambda + \Delta_t)$.

The hand-over. Both calls can return each state's first-order amplitudes: for SC $c_t = -1/(\Delta_t + H^N_t/N^N_t)$ (zero where the guard drops the perturber; η of Eq. (5) in place of H/N for a one-function class, which drops none), so $\Psi^{(1)}_N \ni c_t P_t V |N\rangle$; for PC U_N and $g_{tk} = y_{N,tk}/(\lambda_{N,k} + \Delta_t)$, so $\Psi^{(1)}_N \ni -\sum_k g_{tk} \sum_\mu U_N[\mu, k] \phi^N_\mu$. A later call with its diagonal omitted reads them instead of walking the states again: n states take one diagonal call, then one call per pair over (M, N) and (N, M) alone, holding two transition sets at a time and never n^2. The PT2 transition density (§9.3) contracts the same amplitudes.

4.5 How the pipeline drives it, and the call graph

`pipeline/nevpt2` (`[nevpt2]`) runs only over DF integrals and a closed-shell singlet, and refuses integral-direct ERIs and open shells before anything is queued.

Per root (state-specific, SC and PC from the same inputs): γ_r, Γ_r of the root's CI vector alone; the root's own Fock $F = F^I + F^A(\gamma_r)$, with the core and virtuals canonicalised over it ($C_r = CR$, R block-diagonal), so ε are F's eigenvalues and *not* the CASSCF's state-averaged `mo_energies` (the active orbitals are not rotated to natural orbitals, as PySCF does, since both variants are invariant to that); $R^{\mathsf{T}} F^I R$ sliced into h and the three `h1e_v` blocks; the MO blocks of B over C_r; dm1 $= \gamma_r$, dm2 $= \Gamma_r + \delta_{qr}\gamma_{ps}$, and dm3, f3ca, f3ac by `nevpt2::rdms`; then `class_energies` and `pc_class_energies` over the same inputs. A PC refusal is logged and that root keeps its SC.

With `quasi_degenerate`: one shared set of orbitals, the CASSCF's own, core and virtuals canonical over the state-averaged Fock, its `mo_energies` as ε. So *the QD diagonal differs from the per-root results*: it is root r's NEVPT2 on the shared orbitals, not on its own. Each root's own RDMs are kept from the per-root pass (they do not depend on the core–virtual rotation). Then the diagonal-only call, the per-pair calls over the transition dm1–dm3 with null f3, Eq. (9), and its eigenvectors, each signed so its largest component is positive. A PC-QD refusal leaves PC-QD empty and SC-QD standing.

Figure 1 is `Nevpt2Step::run` as a DAG, from the (SA-)CASSCF's output to the four energies; Figure 2 is what the four entry points call inside the energy layer. Blue nodes are the layer's entry points, orange the RDM builds that dominate as the active space grows. With `quasi_degenerate` the per-root pass still runs first: the QD diagonal reads its RDMs, though not its orbitals.

5 The reverse passes

5.1 One derivative rule

`einsum_adjoint` takes a forward einsum's subscripts, operands, output shape and scale unchanged, and writes operand k's bar as the same einsum with the output and operand k swapping roles. An operand that repeats a label takes its bar over its distinct labels and is scattered onto the diagonal; a strided view takes its bar packed and one GEMM-add places it. Every class lists its norm and H terms once, as data, and both directions read the list, so no derivative of a class is transcribed by hand. The NumPy slice assignments of the reference (`a16[:, i, :, :, :, i] += fdm2` and two in a22) have the gather fdm2^{T}-bar = `einsum("pirabi->prab", a16_bar)` as their adjoint, since an input label may repeat.

5.2 The SC finish

The only hand-differentiated expression on the SC side is Eq. (4)'s finish. With $T_t = -1/(\Delta_t + H_t/N_t)$ [8]:

$$P_t = \frac{\partial E}{\partial N_t} = 2T_t + \Delta_t T_t^2, \qquad Q_t = \frac{\partial E}{\partial H_t} = T_t^2, \qquad R_t = \frac{\partial E}{\partial \Delta_t} = N_t T_t^2, \tag{15}$$

zero for a dropped perturber. A one-function class puts $P_t = c_t$ on N_t, nothing on H_t, $\bar{D}_t = \bar{H}_{JJ} N_t c_t^2$ (plus the QD share) on Δ_t, and $\bar{\eta} = \sum_t \bar{D}_t$ onto its blocks after the walk: $\bar{\eta}\, Y/\langle Y, Y\rangle$ on X and $\bar{\eta}\,(X - 2\eta Y)/\langle Y, Y\rangle$ on Y. P and Q go back through the norm and H einsums onto the slabs (and so Λ), the one-body rows and the blocks' bars $M_G = \partial E/\partial G$; R goes onto the orbital energies Δ_t reads. Sijrs has its own kernel. Srsi walks twice, since its finish pairs entries across slabs.

The sweep. The block bars are swept back through the builders that formed them, in reverse topological order (an intermediate's bar is complete before it is swept), onto $\bar{h}$, the active $\overline{(tu|vw)}$ and $\overline{\mathrm{dm1}}, \ldots, \overline{\mathrm{f3ac}}$. It has two halves: the Dyall blocks, which read h and $(tu|vw)$ and only a state's own walk has, and the metrics, which read only RDMs and are all a QD transition takes. Intermediates are recomputed, never kept.

Identities from the bars. Each class energy is homogeneous of degree one in (N, H), so

$$\sum_t \left(P_t N_t + Q_t H_t\right) = \sum_G \langle M_G, G\rangle = E_k \tag{16}$$

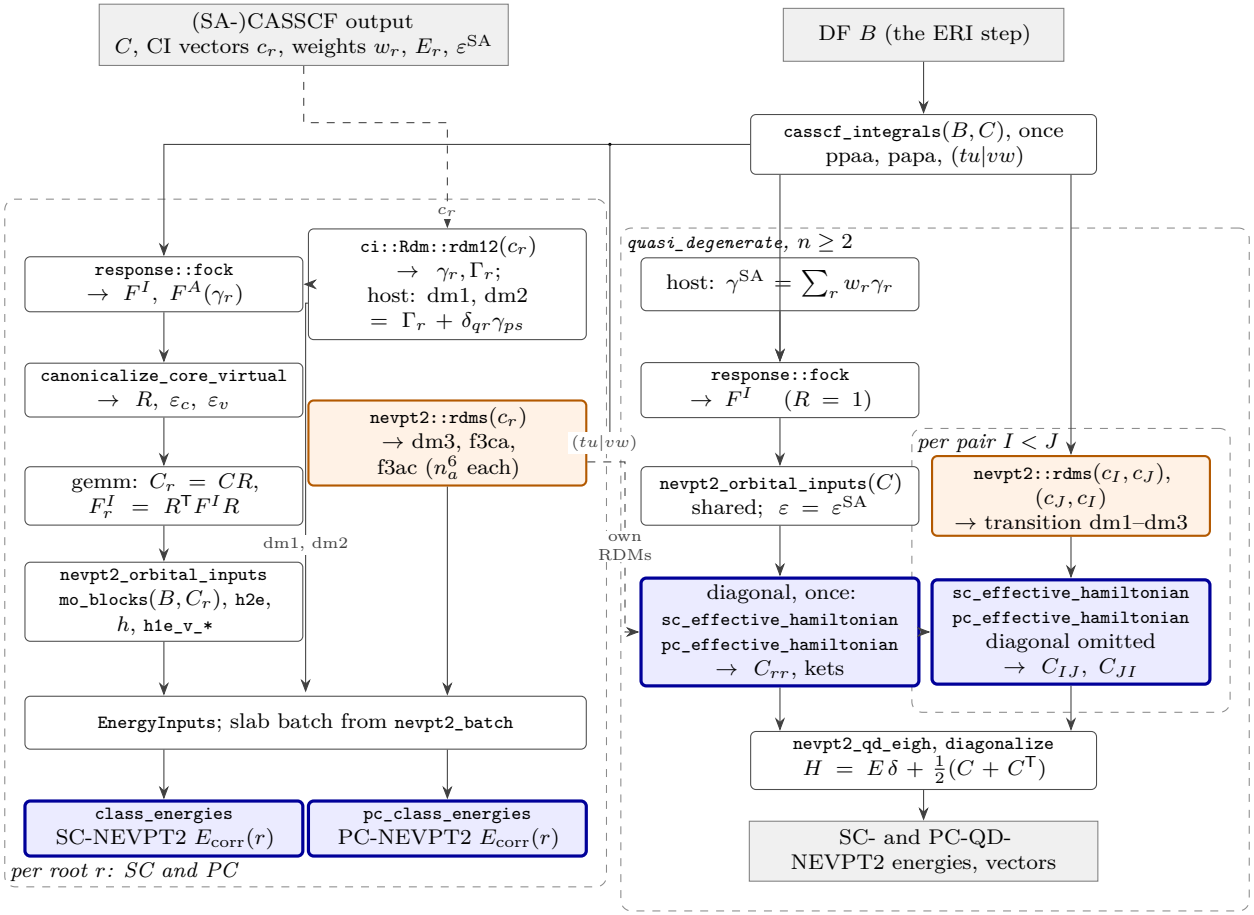


Figure 1: `Nevpt2Step::run` (`src/pipeline/nevpt2/nevpt2.cpp`). Left, every root's own orbitals and RDMs, for SC- and PC-NEVPT2. Right, the shared state-averaged orbitals for QD: the diagonal is walked once and hands its first-order wave functions (the kets) and its inputs to the pair calls, which build only transition RDMs and walk only the transition norms. C and the CI vectors reach every node that names them; dashed edges carry data from the CASSCF or across the two passes.

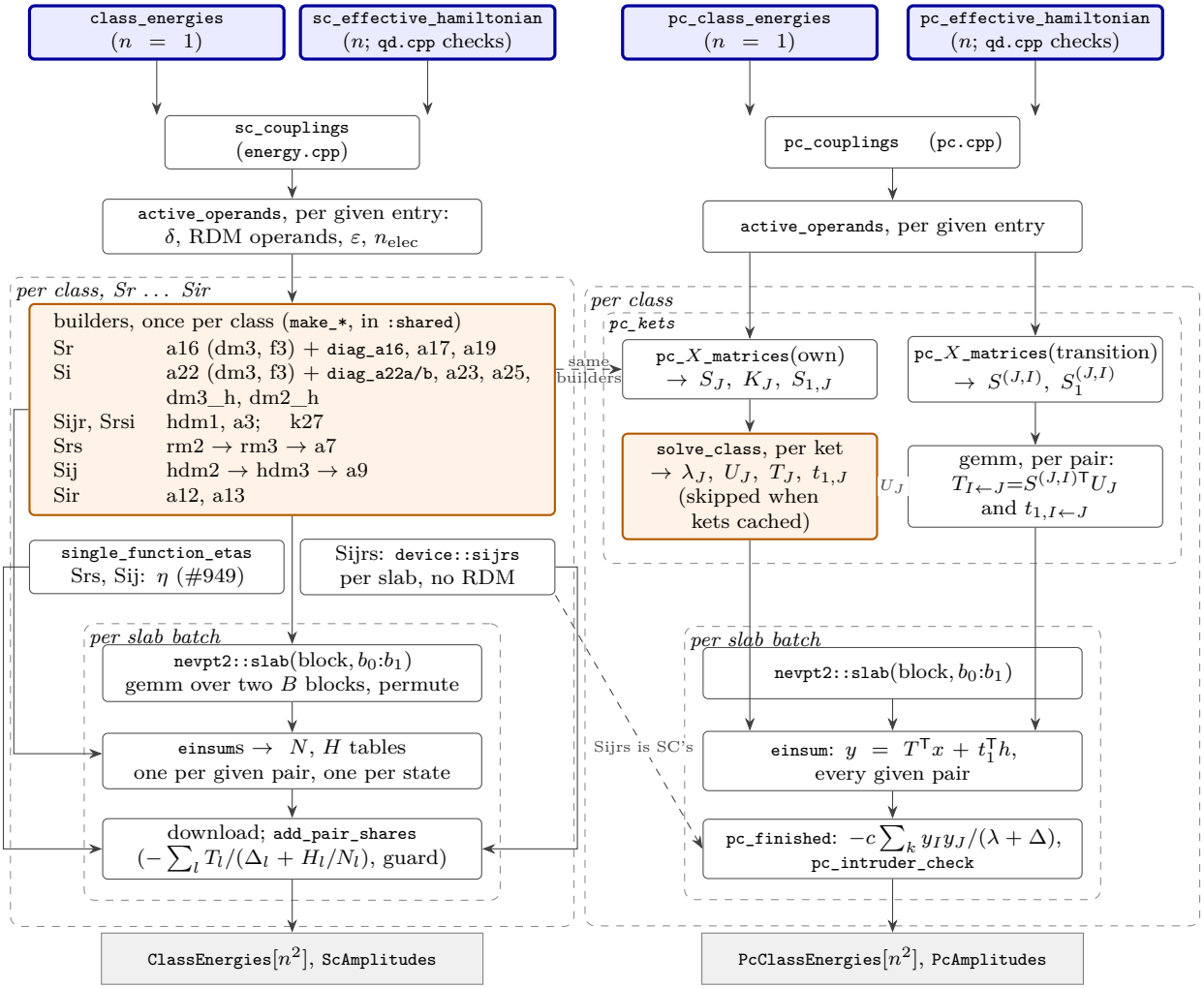


Figure 2: Inside `free_breakfast.nevpt2.energy`. Each single-state call is its effective Hamiltonian's walk at $n = 1$. A class builds its RDM intermediates once, then walks its external index in batches, building one slab of the density-fitted integrals at a time. PC solves each ket's metric once per class (`solve_class`) and reads the slab only through one einsum into that eigenbasis. With the diagonal omitted, both read the kets from the amplitudes a diagonal call returned and solve nothing.

(Park's Eq. 40 [8]); and at fixed orbital energies E is homogeneous of degree 4 in $(B, h_v^{1/2})$, so $\langle \Lambda, B\rangle + 2\sum\langle \bar{h}_v, h_v\rangle = 4E$, h_v the one-body rows. Both are step-free checks of the bars.

5.3 The PC solve, backwards

With $A_t = K + \Delta_t S$ and $x = A_t^+ b_t$ the solve the energy ran,

$$\dot{E}_t = -c(2\, x^\mathsf{T} \dot{b}_t - x^\mathsf{T} \dot{A}_t\, x), \tag{17}$$

so no eigenvalue or eigenvector derivative appears. This holds where S's kept rank is locally constant and K annihilates S's dropped modes, which holds at an eigenstate for its null modes. With real modes dropped at the cut (#853) it is validated at PSB3 against re-converged central differences to max(2× the Richardson bar, $3\sigma_E/h_{\min}$), and keeping those modes does not move it (`src/nevpt2/energy/README.md`). In the eigenbasis, with $g_t = y_t/(\lambda + \Delta_t)$, each tuple gives $\bar{y}_t = -2c\, g_t$, a weight $w_t = c\, g_t$ and $\partial E/\partial \Delta_t = c\, g_t \cdot g_t$; accumulating $M = \sum_t g_t w_t^\mathsf{T}$ and $M_\Delta = \sum_t \Delta_t g_t w_t^\mathsf{T}$ over the walk,

$$\frac{\partial E}{\partial K} = U M U^\mathsf{T}, \qquad \frac{\partial E}{\partial S} = U \bar{\mathcal{T}}^\mathsf{T} + U M_\Delta U^\mathsf{T}, \qquad \frac{\partial E}{\partial S_1} = U \bar{t}_1^\mathsf{T}, \tag{18}$$

symmetrised as the solve symmetrised. Those go back through the einsums that built S, K, S_1 onto SC's block bars, and SC's sweep takes them unchanged. At fixed slabs and orbital energies a PC class is homogeneous of degree one in (S, S_1, K), which gives PC's analogue of Eq. (16).

5.4 The QD reverse passes

Both QD passes differentiate $L = \sum_{MN} \bar{H}_{MN} H_{MN}$ for a given $\bar{H}$; at $n = 1$, $\bar{H} = 1$ they are the single-state passes einsum for einsum.

SC-QD. With $D_N = \Delta + H^N/N^N$, the coupling $-\bar{T}_{MN}/D_N$ has three bars:

$$\frac{\partial L}{\partial \bar{T}_{MN}} = -\frac{\bar{H}_{MN}}{D_N}, \qquad A_N = \sum_M \frac{\bar{H}_{MN}\bar{T}_{MN}}{D_N^2}, \qquad (\bar{N}^N, \bar{H}^N, \bar{\Delta}) \mathrel{+}= \Big(-\frac{A_N H^N}{(N^N)^2},\ \frac{A_N}{N^N},\ A_N\Big), \tag{19}$$

the first split half and half onto the norm einsums of (M, N) and (N, M). The walk runs in three parts: each state's tables, evaluated only; each unordered pair, its norm einsums evaluated and differentiated at once, its metric bars swept onto its transition dm1–dm3 and handed to a callback, then freed; each state's own walk with A_N added. Memory is one pair's transition set at a time, never n^2 sets.

PC-QD. The coupling is bilinear in the two rows, so with $g_M = y_{M\leftarrow N}/(\lambda_N + \Delta)$ and $g = y_N/(\lambda_N + \Delta)$ the bars are $-c\bar{H}_{MN} g$ on $y_{M\leftarrow N}$, $-c\bar{H}_{MN} g_M$ on y_N, $c\bar{H}_{MN} g_M g^\mathsf{T}$ on M_N (and Δ times it on $M_{\Delta,N}$), and $c\bar{H}_{MN} g_M \cdot g$ on Δ. Still no eigenvector derivative: $b^{M\leftarrow N}$ lies in S_N's kept range, since a mode annihilating $|N\rangle$ annihilates $\langle N|S^{(N,M)}$. Then $\partial L/\partial S^{(N,M)} = \bar{\mathcal{T}}_{M\leftarrow N} U_N^\mathsf{T}$ and $\partial L/\partial S_1^{(N,M)} = U_N \bar{t}_{1,M\leftarrow N}^\mathsf{T}$, unsymmetrised, onto pair (N, M)'s transition RDMs. Each class walks its slabs once with every pair, as the forward does.

6 The single-state gradient

6.1 The energy as a function of (h, B, C, c)

`sc_energy` and `pc_energy` rebuild everything the class walk reads from (h, B) in the MO basis of C and the CI vector c:

$$h1e = F^I_{tu}, \quad h1e_v_si = F^I_{ti}, \quad h1e_v_sir = F^I_{ai}, \quad h1e_v_sr = F^I_{at} - \sum_u (au|ut), \quad \varepsilon_x = f_{xx}, \tag{20}$$

dm3, f3ca, f3ac by `rdms`, and dm1, dm2 as dm3's traces (Eq. (13)). Two choices matter:

- ε **moves with** γ**.** It is rebuilt from (h, B, γ) at every point, never an input. Holding it fixed differentiates a different functional, and a check whose two sides share the omission cannot see it.
- **The orbitals are taken as they are.** ε is f's diagonal; off canonical orbitals nothing re-canonicalises. SC-NEVPT2 is not invariant to core–core or virtual–virtual rotations, so the pseudocanonical conditions $f_{ij} = f_{ab} = 0$ belong to the Lagrangian, not the functional.

Beyond the five MO blocks the builders read B_{cc} whole and B_{vv}'s diagonal.

6.2 The adjoints and the orbital source term

(h, B) in the MO basis is the interface because every NEVPT2 input is built from it: the orbital energies, the inactive Fock blocks, the active $(tu|vw)$ and every external slab. So one reverse pass reaches the whole energy, and both the orbital rotations and the nuclear derivative integrals act on (h, B) alone. One reverse pass (§5, then the builders backwards, `rdms_adjoint` and the DF adjoints) gives

$$\bar{h} = \frac{\partial E_{\rm corr}}{\partial h}, \qquad \Lambda^L = \frac{\partial E_{\rm corr}}{\partial B^L}, \qquad y = \frac{\partial E_{\rm corr}}{\partial c}\Big|_{h,B}, \tag{21}$$

$\bar{h}$ and each Λ^L symmetrised last. Along $C \to C(1+\theta)$, any $\Phi(h, B)$ with symmetric adjoints moves by $\sum \theta_{xy} X_{xy}$,

$$X = 2\, h\, \bar{h} + 2 \sum_L B^L \Lambda^L, \qquad A = X - X^{\mathsf{T}} = \frac{\partial \Phi}{\partial \kappa}, \tag{22}$$

X needs B^L_{xq} with x general, which is the full-range B or, without forming it, an AO half-transform:

$$X = 2\, C^{\mathsf{T}} h^{\rm AO} C\, \bar{h} + 2 \sum_L C^{\mathsf{T}} J^L(C\, \Lambda^L), \qquad J^L_{\mu\nu} = \sum_P (L^{-1})_{LP}\, (\mu\nu|P), \tag{23}$$

evaluated as $\tilde{\Lambda}_q = \Lambda_q C^{\mathsf{T}}$ per q (one batched GEMM), then $D = B^{\mathsf{T}}_{\rm AO} \tilde{\Lambda}$, the DF-exchange shape, rather than pair by pair. It is exact by construction: $\langle \kappa, A \rangle$ equals the forward tangent $\dot{E}(\kappa)$, a check rather than an assembly route. Every block of A has a reader: the mixed blocks the Z-vector, core–core and virtual–virtual z^c, active–active the gauge identity. A is Park's orbital source term Y' [8].

6.3 The pseudocanonical multipliers

Along $\kappa_{pq} = -\kappa_{qp} = \theta$ within the core or the virtuals, f_{pq} and f_{qp} move by $\theta(\varepsilon_p - \varepsilon_q)$ and nothing else the constraint reads moves at first order. So stationarity of $E_{\rm corr} + \langle z^c, f \rangle$ (full Frobenius sum,

symmetric z^c) decouples pair by pair:

$$z^c_{pq} = z^c_{qp} = -\frac{A_{pq}}{2(\varepsilon_p - \varepsilon_q)}, \qquad p \neq q \text{ both core or both virtual,} \tag{24}$$

Park's Eqs. 56–57 [8] and Eqs. 25–26 [10] in this tree's sign of κ. `zc_multipliers` takes only A and ε, so it serves SC, PC and QD alike.

Degenerate pairs. For $|\varepsilon_p - \varepsilon_q| < 10^{-6}\,E_h$, a pair with $|A_{pq}| > 10^{-8}$ is refused, naming both orbitals; one with A_{pq} under its rounding bound (n^2_{mo} unit roundoffs of $\max|A|$; partners in different irreps, say) has $z^c_{pq} = 0$; any other keeps Eq. (24). PC's A vanishes with the gap, $A_{pq} = (\varepsilon_p - \varepsilon_q)K_{pq}$ with K_{pq} finite, so PC's z^c has the limit $-K_{pq}/2$ where SC's grows as the inverse gap and is refused: this is Park's reason for preferring PC [8]. Dropping PC's z^c there would be a step of $O(z)$ in the gradient, not $O(\text{gap})$ (#974; `src/nevpt2/gradient/README.md`, "Degenerate pairs").

The term z^c adds. $\Phi = \langle z^c, f\rangle$ with $D = 2$ on the core diagonal and γ (or γ^{SA}) on the active block, $g_L = \sum B^L_{rs} D_{rs}$ and $w_L = \sum B^L_{pq} z_{pq}$:

$$\Phi = \langle z, h\rangle + \sum_L w_L g_L - \tfrac{1}{2}\sum_L \mathrm{tr}(zB^L D B^L), \qquad \bar{h}^{\Phi} = z, \tag{25}$$

$$\Lambda^{\Phi,L} = z g_L + D w_L - \tfrac{1}{2}(zB^L D + D B^L z), \tag{26}$$

$$\mathcal{W} = \frac{\partial \Phi}{\partial \gamma} = \sum_L w_L B^L_{aa} - \tfrac{1}{2}\sum_L (B^L z B^L)_{aa}, \tag{27}$$

$$y^c_r = w_r \sum_{tu} (\mathcal{W} + \mathcal{W}^{\mathsf{T}})_{tu}\, E_{tu} c_r, \tag{28}$$

the last one sigma-vector application of a one-body operator, batched over the roots. y^c is raw (not projected), as y is.

6.4 Lagrangian, Z-vector, gate and nuclear term

On a converged single-state CASSCF,

$$\mathcal{L} = E_{\mathrm{CAS}} + E_{\mathrm{corr}} + Z \cdot g_{\mathrm{orb}} + z \cdot g_{\mathrm{ci}} + \langle z^c, f\rangle, \qquad g_{\mathrm{ci}} = 2(H_0 - E)c, \tag{29}$$

g_{orb} `response::fock`'s orbital gradient over the packed non-redundant pairs. Stationarity in the CASSCF parameters $x = (\kappa_{\mathrm{nr}}, c)$ is the Z-vector equation [13]: with $J = \partial(g_{\mathrm{orb}}, g_{\mathrm{ci}})/\partial x$, the CASSCF electronic Hessian, symmetric at a stationary point,

$$J \begin{pmatrix} Z \\ z \end{pmatrix} = -\begin{pmatrix} A^{\mathrm{corr}}_{\mathrm{nr}} + A^{z^c}_{\mathrm{nr}} \\ P_c(y + y^c) \end{pmatrix}, \qquad z \perp c, \tag{30}$$

E_{CAS} contributing no right-hand side because $g = 0$. Its blocks are what the tree has: $J_{\kappa\kappa}$ `response::Hessian`; $J_{c\kappa}v = 2P_c(H[v] - \langle H[v]\rangle)c$, $H[v]$ the active Hamiltonian at the one-index-rotated integrals; $J_{\kappa c}v$ the orbital gradient at the transition RDMs between v and c; and $J_{cc}v = 2P_c(H - E_{\mathrm{CAS}})P_c v$, one sigma vector. It is solved by `response::zvector` (preconditioned CG on `recipes::pcg`, CI block projected off c before and after the preconditioner, since c is an exact null vector of J_{cc}): orbital right-hand side $g + \mathrm{pack}(A^{\mathrm{corr}} + A^{z^c})$ and CI right-hand side $\frac{1}{2}P_c(y + y^c)$. The $\frac{1}{2}$ is `zvector`'s convention: its CI row is half of Eq. (30)'s, so a caller of Eq. (30)'s form scales its CI right-hand side to match.

The total adjoint. Every term of Eq. (29) but V_{nn} is a function of (h, B) at fixed (c, Z, z, z^c), so the nuclear term reads the sum of their adjoints,

$$\bar{h}^{\mathcal{L}} = \bar{h}^{\text{cas}} + \bar{h}^{\text{corr}} + \bar{h}^{Z} + \bar{h}^{z} + \bar{h}^{z^c}, \qquad \Lambda^{\mathcal{L}} = \Lambda^{\text{cas}} + \Lambda^{\text{corr}} + \Lambda^{Z} + \Lambda^{z} + \Lambda^{z^c} : \tag{31}$$

CAS, $\bar{h}^{\text{cas}} = D$ and $\Lambda^{\text{cas},L}_{pq} = \sum_{rs} \Gamma_{pqrs} B^L_{rs}$ (the core parts J/K shapes, the active part $\Gamma^{(2)}$ against B_{aa}); Z, E_{CAS}'s adjoints at the rotated densities $D^Z = Z^{\mathsf{T}} D + D Z$ and their four-index analogue; z, E_{CAS}'s at the symmetrised transition densities between z and c, times 2; z^c, Φ's above; corr, the reverse pass.

The gate: a free, exact check of the whole solve. $\mathcal{L}$ is stationary in every rotation, so by Eq. (22) its total $X^{\mathcal{L}}$ must be symmetric in every block, gauge blocks included. Its antisymmetric part,

$$A^{\mathcal{L}} = 2(F - F^{\mathsf{T}})(\gamma + \gamma_z, \Gamma + \Gamma_z) + A^{Z} + A^{\text{corr}} + A^{z^c}, \tag{32}$$

with F the generalised Fock at the root's RDMs plus z's transition RDMs, and $A^Z = 2(F' - F'^{\mathsf{T}})$, F' the derivative of F along $C(1 + \epsilon\kappa(Z))$, is evaluated and every block past 10^{-8} is refused, naming it. F is a polynomial of degree 4 in C, so a five-point stencil gives F' exactly to round-off. On the non-redundant pairs $A^{\mathcal{L}}$ is the Z-vector's residual, on the inactive pairs Eq. (24)'s, on the active pairs the gauge identity's: a wrong factor in any term names its block.

The nuclear term. Along any orthonormality-preserving path the symmetric part of θ is $-\frac{1}{2} C^{\mathsf{T}} S^A C$, so with the total adjoints $(\bar{h}^{\mathcal{L}}, \Lambda^{\mathcal{L}}, X^{\mathcal{L}})$,

$$\frac{dE}{dR_A} = \sum_{\mu\nu} \mathcal{P}_{\mu\nu} h^A_{\mu\nu} + \sum_{\mu\nu P} \mathcal{G}^P_{\mu\nu} (\mu\nu|P)^A - \tfrac{1}{2} \sum_{PQ} \mathcal{V}_{PQ} (P|Q)^A - \sum_{\mu\nu} \mathcal{W}_{\mu\nu} S^A_{\mu\nu} + V^A_{nn}, \tag{33}$$

$\mathcal{P} = C \bar{h}^{\mathcal{L}} C^{\mathsf{T}}$, $\mathcal{G}^P = \sum_L (L^{-\mathsf{T}})_{PL} C \Lambda^{\mathcal{L},L} C^{\mathsf{T}}$, $\mathcal{V} = L^{-\mathsf{T}} \mathcal{G}^{\text{aux}} L^{-1}$ with $\mathcal{G}^{\text{aux}}_{LL'} = \langle \Lambda^{\mathcal{L},L}, B^{L'} \rangle$, and $\mathcal{W} = \frac{1}{2} C \,\text{sym}(X^{\mathcal{L}}) C^{\mathsf{T}}$. In the code this is one `mcscf::df_gradient` pass: E_{CAS}'s and the multipliers' terms are the ones it builds, and $(\bar{h}, \Lambda, X)$ of $E_{\text{corr}} + \Phi$ are its "extra" MO adjoint. No relaxed density is formed separately.

Algorithm 2 The DF-NEVPT2 nuclear gradient (`sc_gradient`, `pc_gradient`)

Input: converged single-state DF-CASSCF (C, c), core and virtuals canonical; SC or PC
Output: dE/dR

1: $(E_{\text{corr}}, \bar{h}, \Lambda, y, X, A, \varepsilon) \leftarrow$ `sc_adjoint` or `pc_adjoint`
2: $z^c \leftarrow$ Eq. (24); refuse an undetermined degenerate pair
3: $(\Phi, \bar{h}^{\Phi}, \Lambda^{\Phi}, y^c, X^{\Phi}, A^{\Phi}) \leftarrow$ `zc_adjoint`; add field by field
4: $(Z, z) \leftarrow$ `zvector`$\big(g + \text{pack}\, A,\ \frac{1}{2}(y + y^c)\big)$
5: **gate:** $A^{\mathcal{L}}$ below 10^{-8} in all six blocks
6: $dE/dR \leftarrow$ `df_gradient`$\big(\text{multipliers } (Z, z),\ \text{extra } (\bar{h}, \Lambda, X)\big)$

7 The single-state nuclear Hessian

`hessian::df_sc_nevpt2` and `hessian::df_pc_nevpt2` (#816, #817) return $d^2(E_{\text{CAS}} + E_{\text{corr}})/dR_A\, dR_B$ on a converged single-state DF-CASSCF, in k coordinates $q_i = \sum_x Q_{ix} R_x$ (orthonormal rows, the Cartesian ones when none are given). They are all-electron, and a state

average or a frozen core is refused. They are assembled in `hessian` from this layer's tangents and Hessian-vector products and from the CASSCF Hessian's parts, with no integrals of their own. The QD-NEVPT2 Hessian is §11.

7.1 The $2n+1$ form

Collect the pieces of Eq. (29): the parameters $x = (\kappa_{\rm nr}, \kappa_{oo}, c)$, the non-redundant CASSCF rotations, the core–core and virtual–virtual rotations the pseudocanonical condition pins, and the CI vector; the multipliers $\lambda = (Z, z, z^c)$; the constraints $g = (g_{\rm orb}, g_{\rm ci}, f_{oo})$. At the solution $\mathcal{L}_x = 0$ and $g = 0$, so $dE/dR_A = \mathcal{L}_A$ at fixed (x, λ) along an orthonormal path. Differentiating once more and eliminating λ_B with the two first-order conditions, $g_A + g_x x_A = 0$ and $\mathcal{L}_{xB} + \mathcal{L}_{xx} x_B + g_x^{\mathsf{T}} \lambda_B = 0$,

$$\frac{d^2E}{dR_A\, dR_B} = \mathcal{L}_{AB} + \mathcal{L}_{Ax}\, x_B + x_A^{\mathsf{T}} \mathcal{L}_{xB} + x_A^{\mathsf{T}}\, \mathcal{L}_{xx}\, x_B, \qquad g_x\, x_A = -g_A. \tag{34}$$

The parameters are needed to first order, one linear solve per coordinate. The multipliers are the gradient's own, at zeroth order, so their $3N$ responses never appear, and no second-order response x_{AB} does either. SC's amplitudes are closed form and PC's come from a solve whose factorisation the tangent reuses, so neither needs an amplitude response of its own.

7.2 The parameter response and the tangent

g_x is block triangular: the CASSCF constraints transform covariantly under κ_{oo} and do not respond to it at a stationary point.

1. **CP-CASSCF**, $J(\kappa_{\rm nr}, c)_A = -\partial g_{\rm CAS}/\partial R_A$, J the gradient's CI-relaxed Hessian: `hessian::df_casscf`'s step (or `df_casscf_cp` alone), `response::zvectors` on all $3N$ columns at once, CI right-hand sides halved to the Z-vector's convention. A state-specific excited root is a saddle in (κ, c) and is refused (Divergence); so is a singular $\tilde{H}$, detected by a probe column (#978).

2. **The pseudocanonical block**, closed form once 1. is done:

$$\kappa^A_{pq} = -\frac{f^{(A)}_{pq}}{\varepsilon_p - \varepsilon_q}, \qquad (p, q) \text{ core–core or virtual–virtual}, \tag{35}$$

   $f^{(A)}$ the total first-order change of f at fixed κ_{oo}. Below $|\varepsilon_p - \varepsilon_q| = 10^{-6}$, $\kappa^A_{pq} = 0$ while $|f^{(A)}_{pq}| \le 10^{-8}$ (a split the displacement does not lift); past that SC refuses, naming the pair and the coordinate, and PC records a degenerate pair, which its Hessian refuses (§7.5).

Eq. (35) is z^c's division again, now in a response that enters Eq. (34) quadratically. PC does not escape it: this tree's PC puts each tuple at its own Δ_t from f's diagonal and is not invariant to inactive rotations, so κ_{oo} stays in PC's Hessian too.

`hessian::df_nevpt2_tangents` then forms the total first-order change of the NEVPT2 functional's inputs along each coordinate, in the MO basis:

$$\dot{h}^x = h^x + \kappa^{x\mathsf{T}} h + h\kappa^x, \qquad \dot{B}^x_Q = W^x_Q + \kappa^{x\mathsf{T}} B_Q + B_Q \kappa^x, \qquad \dot{c}^x = c^x, \qquad \kappa^x = \kappa^x_{\rm nr} + \kappa^x_{oo}, \tag{36}$$

(h^x, W^x) the derivative integrals along the Löwdin connection $\tau^x = -\frac{1}{2} S^x_{\rm MO}$ (`mcscf::df_integral_derivatives`), W^x's fitted index carried in the gauge $\mu^x = -\frac{1}{2} m^x$, $m^x = L^{-1} M^x L^{-\mathsf{T}}$. $f^{(x)}$ comes from `nevpt2::builders_tangent`, the forward tangent of every builder the energy reads, each term by its product rule over the forward's own builds. The RDMs' tangent is exact in three `rdms` calls: the bra/ket builds at $(\dot{c}, c)$ and $(c, \dot{c})$, and the one-vector build over $(tu|vw)$'s tangent, of which only f3 is kept.

7.3 The Hessian-vector product

`sc_hessian_product` and `pc_hessian_product` (#814, #815) are the tangent of `*_adjoint` + `zc_adjoint` at fixed z^c along directions $(\dot{h}, \dot{B}, \dot{c})$, so $\langle t_A, \mathrm{HVP}\, t_B \rangle$ is the $E_{\mathrm{corr}} + \Phi$ share of Eq. (34) along Eq. (36)'s t. The adjoint is $J^{\mathsf{T}}\bar{E}$, $\bar{E}$ the class walk's bars at the builders' outputs and J the builders' Jacobian, so its tangent is forward over reverse, two kinds of term, one sweep (`bars_tangent`):

- $J^{\mathsf{T}}\dot{\bar{E}}$: the class walk's bars along the builders' tangent (`class_energies_hvp`, `pc_class_energies_hvp`), run back through the gradient's own reverse pass;
- $\dot{J}^{\mathsf{T}}\bar{E}$: every builder term's adjoint at the fixed bars with one other operand at its tangent, the same einsum with the operand swapped (`adjoint_slot`), so no derivative is written by hand.

SC's class tangent. The multipliers' tangent is the finish's 3×3 Hessian. In the (N, d) form, $d = N\Delta + H$ and $E = -N^2/d$, which carries no removable $1/N$:

$$\begin{aligned} E_{NN} &= -\frac{2}{d} + \frac{4N\Delta}{d^2} - \frac{2N^2\Delta^2}{d^3}, & E_{NH} &= \frac{2N}{d^2} - \frac{2N^2\Delta}{d^3}, & E_{N\Delta} &= \frac{3N^2}{d^2} - \frac{2N^3\Delta}{d^3}, \\ E_{HH} &= -\frac{2N^2}{d^3}, & E_{H\Delta} &= -\frac{2N^3}{d^3}, & E_{\Delta\Delta} &= -\frac{2N^4}{d^3}, \end{aligned} \tag{37}$$

whose (H, Δ) block has rank one, and whose degree-one homogeneity in (N, H) gives $NE_{NN} + HE_{NH} = 0 = NE_{NH} + HE_{HH}$. Times the forward tangent's $(\dot{N}, \dot{H}, \dot{\Delta})$ it gives $(\dot{P}, \dot{Q}, \dot{R})$, taken onto every bar by one reverse pass at fixed multipliers. The bars at fixed multipliers are themselves polynomials of degree at most five along a line (two slabs, each quadratic in B, around a block of degree at most two), so

$$\sum_{k=1}^{3} w_k \, \frac{A(x + ks\dot{x}) - A(x - ks\dot{x})}{2ks}, \qquad w = (\tfrac{3}{2}, -\tfrac{3}{5}, \tfrac{1}{10}), \tag{38}$$

is their derivative exactly, the weights cancelling the s^2 and s^4 terms. A one-function class reads $\dot{\eta}$ from the blocks' polarisation.

PC's class tangent. The same split over the solve. With $g = y/(\lambda + \Delta)$ the multipliers move as $\dot{g} = (\dot{y} - \dot{D}g)/(\lambda + \Delta)$, $\dot{D} = \dot{K}' + \Delta\dot{S}' + \dot{\Delta}$, the solve differentiated through its own factorisation. U's tangent $V_n\Theta'$ enters through a symmetric difference that is exact (the bars are quadratic in U), the inputs' tangent at fixed multipliers through the same three-point formula, and the kept span's rotation through the Daleckii–Krein derivative of the span bar. Every divided difference there is a product of two first-order ones, so no pair on one side of the cut is ever divided by. These are the null-space rotation terms of $S^+ = UU^{\mathsf{T}}$: zero for exact null modes at an eigenstate, and of the first-order term's size for real modes near the cut.

Φ, and the functional of c. $\Phi = \langle z^c, f \rangle$ is linear in h, bilinear in B and quadratic in c, so its share at fixed z^c is closed form. Two conventions fix the CI–CI block, and the gradient shares both. The product is the Hessian of the *Rayleigh-normalised* functional $F(h, B, c) = (E_{\mathrm{corr}} + \Phi)(h, B, c/|c|)$, so the sphere's $-(u \cdot v)\,(\partial\mathcal{L}/\partial c) \cdot c$ is included: with $P = 1 - cc^{\mathsf{T}}$ and $u = \dot{c} - (c \cdot \dot{c})c$, the c-gradient's tangent is $P\dot{y}(\dot{h}, \dot{B}, u) - u\,(c \cdot y) - c\,(u \cdot y) - (c \cdot \dot{c})Py$. The off-shell extension is the rank-reduced one, $\hat{H}_D|\Psi_0\rangle = E_0|\Psi_0\rangle$ off the root too, with no Kollmar residual [12]. A residual-including CI–CI block against this gradient's z, or the reverse, would be wrong with no symptom.

7.4 Assembly

Every part is the second derivative of its term of $\mathcal{L}$ along one path: the orbitals $C(R)\exp K$ along the Löwdin connection $C(R) = C(C^{\mathsf{T}}S(R)C)^{-1/2}$, $K = \kappa_{\mathrm{nr}} + \kappa_{oo}$, the CI vector $\exp(A)c$ on the unit sphere, and B's fitted index in one gauge. A part in another parametrization would leave a term of second order in κ unaccounted. With $N = E_{\mathrm{corr}} + \Phi$ a function of $I = (h, B)$ and c, $O(M,\kappa) = \kappa^{\mathsf{T}}M + M\kappa$ and $t_A = I^A + O(I,\kappa^A) + c^A$:

$$\begin{aligned} N^{\mathrm{total}}_{AB} = {} & \langle N_I, I^{AB}\rangle + \langle t_A, N''\, t_B\rangle \\ & + \langle N_I, O(I^A,\kappa^B) + O(I^B,\kappa^A)\rangle + \langle N_I, \tfrac{1}{2}O(O(I,\kappa^A),\kappa^B) + (A \leftrightarrow B)\rangle, \end{aligned} \tag{39}$$

the skeleton, the product, and the rotation cross terms $\tau\kappa$ and $\kappa\kappa$. For symmetric M, $\langle N_I, O(M,\kappa)\rangle = \langle \kappa, X(M)\rangle$, $X(M) = 2(M_h\bar{h} + \sum_Q M_Q\Lambda_Q)$, so the cross terms are n^2_{mo} Gram matrices. The parts and what computes each:

part	computed by
E_{CAS}	`hessian::df_casscf` as is; its κ_{oo} rows vanish by $E_{\mathrm{CAS}}(CU,c) = E_{\mathrm{CAS}}(C,c)$
t_x	`df_nevpt2_tangents` over its λ^x, with κ^x_{oo}
$(\bar{h},\Lambda)$, multipliers	`*_adjoint`, `zc_multipliers`, `zc_adjoint`, `lagrangian_gradient`: the gradient's chain, adjoint kept
$\langle t_A, N'' t_B\rangle$	`*_hessian_product` along each coordinate's t, paired with every t_A by three GEMVs
Z terms	`df_casscf_z`: $Z\cdot\partial^2 g/\partial R\,\partial R$, $I^x Z$ against λ^y both ways, and $\lambda^{\mathsf{T}}(Z\cdot\partial^2 g)\lambda$, the CASSCF third derivative
skeleton	`df_adjoint_xy`: $\sum \bar{h}\, h^{xy} + \sum \Lambda\, B^{xy}$
$\tau\kappa$, $\kappa\kappa$	X Gram matrices from n^2_{mo} stacks
κ_{oo} rows	$\frac{1}{2}\big(\langle \kappa^B_{oo}, [X_N, \kappa^A_{\mathrm{nr}}]\rangle + (A \leftrightarrow B)\big)$

The κ_{oo} rows follow from BCH to second order, $\exp(sP{+}tQ) = \exp(sP)\exp(tQ)\exp(-\frac{1}{2}st[P,Q]){+}O(3)$, and the covariance of the orbital gradient: of $Z\cdot g$'s second derivative only $-\frac{1}{2}\nabla(Z\cdot g)\cdot([\kappa^A_{\mathrm{nr}},\kappa^B_{oo}] + [\kappa^B_{\mathrm{nr}},\kappa^A_{oo}])$ survives, and the Z-vector equation turns $\nabla(Z\cdot g)$ into $-X_N$. N's own κ_{oo} terms are in t's κ, so the product and the cross terms carry them.

The skeleton's gauge. h_{MO} and B_{MO} are differentiated along $\tau^{xy} = -\frac{1}{2}S^{xy} + \frac{3}{8}\{S^x, S^y\}$ and, on the fitted index, $\mu^{xy} = -\frac{1}{2}m^{xy} + \frac{3}{8}\{m^x, m^y\}$: the symmetric square root's second order, whose first order is exactly W^x's gauge. The fitted index is defined only up to an orthogonal turn, which a general Λ sees; the product's term pairs W^x and this one B^{xy}, and their sum is the derivative of one function only because both are the same curve.

The result is exactly symmetric; the antisymmetric part discarded and the translational sum are returned.

7.5 Where there is no Hessian: the refusals

Each refusal below is Numerical, names what failed, and returns no Hessian; the CP solve's own (a saddle, a singular $\tilde{H}$) are passed on as they come.

SC: empty perturbers. At a symmetric geometry a perturber can be empty by symmetry ($|N| \le 10^{-14}$); a direction that breaks the symmetry populates it at second order, where its energy is homogeneous of degree two in the direction. Along one direction u it is a quadratic form, and the perturber takes the demo's K6, the finish's multipliers at the ray's limit $\eta = H^{uu}/N^{uu}$.

Along a pair (u, v) that both populate it, it takes K6 at the mean η and is scored by $\rho_\tau = \max(|\eta_u - \eta_v|, 2|\kappa - c\eta|)/|\Delta + \eta|$ at unit populations, zero exactly when $H \propto N$ on the pair, weighted by the perturber's share of the pair's Hessian (#914). Past 10^{-6} the pair is refused: the energy is a second derivative along every line but no bilinear form, #894's SC kink. Because the product is not linear in the direction there, the Hessian is taken along each coordinate's own t and pairs of symmetry-breaking coordinates are taken again as each other's partners. A coordinate breaks the symmetry when its population exceeds 10^{-10} (`kEmptyPopulationFloor`) of the *largest* scale over the coordinates (#952: a rigid translation's tangent is round-off and populates everything against its own scale). At `nh3_cs_sto3g` in symmetry-adapted coordinates the whole $A'' \times A''$ off-diagonal block is refused, which is the right output.

SC: mixing inactive pairs (#952). SC is not invariant to inactive rotations. Where a coordinate couples two near-degenerate inactive orbitals, the pair's 2×2 block $[[\varepsilon_p, Ft], [Ft, \varepsilon_q]]$ has gap $\sqrt{g^2 + 4F^2t^2}$, and its avoided crossing has half-width $t_{\text{mix}} = g/2|F| = 1/(2|\kappa^i_{oo,pq}|)$. Past it the pair has rotated and the curvature is a different branch's. At `examples/nevpt2.toml`'s C_{2v} water, virtuals $1.4\,\text{m}E_h$ apart give $t_{\text{mix}} = 5.6 \times 10^{-4}$ bohr along B_2, and SC's B_2 frequency is 14063 against PC's $4028\,\text{cm}^{-1}$. The analytic element is the correct $t \to 0$ limit, which a ladder of the gradient reaches only below 10^{-4} bohr. Any coordinate with $t_{\text{mix}} < 10^{-3}$ bohr is refused, naming the pair and recommending PC.

PC: no partners, but the Schur lift. PC has no empty perturbers: a tuple that vanishes by symmetry has $b = Sx = 0$ and moves as a smooth quadratic. Instead a null mode v of S can rise at second order along the CI vector's path. With $S(t) = S + tS' + \frac{1}{2}t^2S''$ the null block's eigenvalues rise as $\frac{1}{2}t^2\,\text{eig}(L)$,

$$L = \mathcal{P}\,(S'' - 2S'S^+S')\,\mathcal{P}, \tag{40}$$

$\mathcal{P}$ the null projector, $L \succeq 0$. Structural null modes never lift ($L = 0$), spin ones not along singlet directions, and symmetry ones do along a singlet that breaks the symmetry. A lifted mode crosses the cut at $t^* = \sqrt{2\,\text{cut}\,\sigma_{\max}/\ell}$, $\ell = \text{eig}(L)$, beyond which the energy carries its share. The product stays the exact derivative of the dropped branch and reports t^* per direction and class. Along the coordinate's own t (its CI part c^B) t^* is in bohr, and any coordinate and class with $t^* < 10^{-3}$ bohr is refused. HCN's CAS(6,6) with its σ pair active lifts Sir's modes at $2\text{–}5 \times 10^{-4}$ bohr on every bend; at $f = 10^{-10}$, t^* grows to 5×10^{-3}–10^{-2} and nothing is refused.

PC: thin kept spectra and degenerate pairs. A class whose `kept` is below 10^{-10} of its $\sigma_{\max}$ is refused, naming the floor that would cut it (#917; #943 reaches it at a C_{2v} formaldehyde CAS(6,6)). A degenerate inactive pair that a coordinate splits (gap below $10^{-6}\,E_h$, $|f^{(x)}_{pq}| > 10^{-8}$) is refused, since κ^x_{oo} is not a division the state determines there. The check is conservative: any Cartesian record refuses, even if the caller's coordinates combine them so that f vanishes.

PC near a degenerate pair, measured. `h2co_neardeg_sto3g`'s virtual pair is $6 \times 10^{-3}\,E_h$ apart, $\max|\kappa^x_{oo}| = 21.8$, and its reference ladder runs through gaps of 5.6×10^{-5}. The terms that grow as $1/\text{gap}^2$, $\kappa\kappa$ and the product's κ_{oo}–κ_{oo} block, are each about 10^9 bars and cancel to the fixture's bar: about nine of the parts' sixteen digits go into it. So PC's second-order κ_{oo} terms do cancel the $1/(\varepsilon_p - \varepsilon_q)$ of Eq. (35), at the cost of digits, and that pair is not refused.

7.6 The semi-numerical Hessian and the bilinearity check

The numerical alternative differences the analytic gradient,

$$H_{AB} \approx \frac{g_A(R + s\,q_B) - g_A(R - s\,q_B)}{2s}, \qquad H \leftarrow \tfrac{1}{2}(H + H^{\mathsf{T}}), \tag{41}$$

along each coordinate q_B, in the pipeline by a fourth-order stencil. Each displaced CASSCF starts from the reference's orbitals orthonormalised in the displaced metric, $C(S_{\mathrm{MO}})^{-1/2}$, and its CI, and passes the CASSCF chain's continuity guards. It is also the oracle for the analytic Hessian. It does not escape the differentiability problem of §7.5: along a pair of directions that leave the symmetry, a difference converges, if at all, to a polarisation of directional second derivatives that depends on the step and on the combination of coordinates.

The diagnostic. For coordinates q_1, q_2, check

$$H(q_1 + q_2,\ q_1 + q_2) \stackrel{?}{=} H(q_1, q_1) + H(q_2, q_2) + 2H(q_1, q_2), \tag{42}$$

with the step dependence of each element. A violation above the difference noise means that block has no Hessian, and it is reported rather than published as a frequency. With one coordinate in an irrep Eq. (42) is empty and the ladder is the only evidence.

Which coordinates. Across two irreps the cross block vanishes by symmetry in symmetry-adapted coordinates, but the function need not be a quadratic form there either. A Cartesian step mixes irreps, so a Cartesian difference Hessian at a symmetric point depends on the frame, silently. Symmetry-adapted coordinates avoid that and name the failing block; on Cartesian pairs the check still detects a failure, at two more gradients per pair, without naming the irrep. The tree detects no point group, so the coordinates are the caller's (`[hessian] coordinates`, orthonormal rows). Element (i, j) is $q_i^{\mathsf{T}}\,\partial g/\partial q_j$, and a Cartesian Hessian is $Q^{\mathsf{T}}HQ$.

In the pipeline (#812). Under `[nevpt2_gradient]` the numerical `[hessian]` runs a ladder by default: each column at $s = h$, $h/2$, $h/4$, the ratio of successive changes judged in (12.8, 19.2) where the finer change is past ten noise floors, the floor measured from restarts and from resolves at tolerances a hundred times tighter. It checks Eq. (42) on every coordinate pair up to twelve coordinates. A failing pair withholds the frequencies: the run fails, naming the pairs. At `nh3_cs_sto3g` under SC in symmetry-adapted coordinates all ten A″×A″ pairs fail and the cross-irrep pairs pass, which is the block the analytic route refuses. Algorithm 3 is the whole method.

7.7 In the pipeline

`[hessian]` beside a single-state `[nevpt2_gradient]` is that variant's $E_{\mathrm{CAS}} + E_{\mathrm{corr}}$ Hessian (#818): analytic by default, or central differences of `sc_gradient`/`pc_gradient` over $6N$ displaced geometries under a Richardson ladder with a bilinearity check (#812). `coordinates`, when given, are passed through and the result reported in Cartesians, $Q^{\mathsf{T}}HQ$: at a symmetric geometry give the symmetry-adapted ones. Under `variant = "pc"` the whole chain runs at `pc_floor` $= 10^{-10}$ (`HessianStep::kPcFloor`), the outer `[nevpt2_gradient]` included, so the gradient written beside the Hessian is the same energy's; `pc_floor = 0` restores the round-off cut bitwise. Refused before anything runs: integral-direct ERIs, a state average (outside a QD chain), a frozen core and a `[casscf_gradient]` beside it. Beside a `quasi_degenerate [nevpt2_gradient]` the numerical method is the semi-numerical QD Hessian (#959, §13), and the analytic one is refused until #968.

Algorithm 3 The semi-numerical NEVPT2 Hessian (`pipeline/hessian`'s numerical method)

Input: the reference geometry's converged CASSCF and NEVPT2 gradient; coordinates q_i (the caller's orthonormal rows, or Cartesian); step h

1: **for** each coordinate q_B, at $s = h, h/2, h/4$ under the ladder (the default; without it $s = h$ alone) **do**
2: **for** each displacement $\pm s\,q_B$, $\pm 2s\,q_B$ (without the ladder $\pm h\,q_B$) **do**
3: re-converge the CASSCF from the reference's orbitals orthonormalised in the displaced metric, $C(S_{\mathrm{MO}})^{-1/2}$, and its CI ▷ warm start
4: continuity guards on every root, refusing on failure; for a QD state, track it by $|\langle R_P|u_k\rangle|$ (§13)
5: the analytic NEVPT2 gradient (one Z-vector solve)
6: column $H_{\cdot B}$ at s: the fourth-order central form of Eq. (41) (without the ladder, Eq. (41) itself)
7: under the ladder, judge each element's ratio of successive changes in $(12.8, 19.2)$ past ten noise floors; the finest rung's Hessian
8: symmetrise; report the asymmetry
9: Eq. (42) on every coordinate pair, up to twelve coordinates; a failing pair withholds the frequencies
10: $Q^{\mathsf{T}}HQ$; harmonic analysis

Stationarity is assumed, not checked: converge `[casscf]` to `conv_tol_grad` $= 10^{-10}$. Harmonic analysis follows; there are no IR intensities.

8 The QD-NEVPT2 gradient

8.1 The functional $\mathcal{E}[W]$

For any symmetric $n \times n$ W,

$$\mathcal{E}[W] = \sum_{MN} W_{MN} H^{\mathrm{eff}}_{MN} = \sum_{M} \rho_M E_M + \sum_{MN} W_{MN} C_{MN}, \qquad \rho_M = W_{MM}, \tag{43}$$

and state K's energy is $E_K = \mathcal{E}[u_K u_K^{\mathsf{T}}]$ ($u_K = R_K$; Park's R_{PPMN} [10]). R_K is an eigenvector of a symmetric matrix, so $\delta E_K = u_K^{\mathsf{T}} \delta H^{\mathrm{eff}} u_K$: R needs no multiplier, and the gradient of E_K is that of $\mathcal{E}[W]$ at fixed W. Against a symmetric W the symmetrisation drops out, so the unsymmetrised couplings are what is differentiated. `qd_sc_energy` and `qd_pc_energy` build every input from (h, B, C) and the roots at their SA weights: one set of shared arrays, ε from f^{SA} with γ^{SA} rebuilt from the roots at every point, and each ordered pair's dm1–dm3.

The reverse pass. `qd_*_adjoint` runs the QD reverse pass at $\bar{H} = W$, hands each transition's dm1–dm3 bars (folded into dm3 along Eq. (13)) to the bra/ket `rdms_adjoint` with null f3 bars, then the shared builders backwards, so that $\bar{\gamma}^{\mathrm{SA}}$ collects in one bar. Each root then takes $w_r \bar{\gamma}^{\mathrm{SA}}$, whatever its W_{rr}: every root carries ε's share through γ^{SA}.

8.2 The Lagrangian on a state average

The SA-CASSCF parameters are the non-redundant rotations, the roots c_r and the rotations Θ among them. With (Z, z, Θ) in `response::zvector`'s coordinates,

$$\mathcal{L} = \mathcal{E}[W] + Z \cdot g_{\mathrm{SA}} + \sum_r 2w_r\, z_r \cdot (H - E_r)c_r + \sum_{r<s} \big(2(w_r - w_s)\Theta_{sr} + \zeta_{sr}\big)\langle c_s|H|c_r\rangle + \langle z^c, f^{\mathrm{SA}}\rangle, \quad (44)$$

Park's Eq. 24 [10] made explicit about the rotations among the roots: a pair of unequal weights is a row of the SA Hessian (Θ), and a pair of equal weights, to which E_{SA} is blind, needs a multiplier ζ of its own. Unlike a CASSCF root energy, C_{MN} is not stationary under mixing the references, which is QD-NEVPT2's reference-basis dependence [7]. Along $c_r \to c_r + \theta c_s$, $c_s \to c_s - \theta c_r$,

$$\tau_{sr} = c_s^{\mathsf{T}}\bar{y}_r - c_r^{\mathsf{T}}\bar{y}_s, \qquad \zeta_{sr} = -\frac{\tau_{sr}}{E_s - E_r}\ (w_r = w_s), \qquad t_{sr} = \frac{\tau_{sr}}{2(w_r - w_s)}\ (w_r \neq w_s), \quad (45)$$

$\bar{y}_r$ the raw CI bar of $\mathcal{E}^{\mathrm{corr}}[W] + \Phi^{\mathrm{SA}}$. ζ's orbital term is $\zeta_{sr} g^{(sr)}$, the orbital gradient of $\langle c_s|H|c_r\rangle$ (`response::fock` at the transition RDMs less at zero RDMs), and its nuclear term is the Hellmann–Feynman gradient of $\langle c_s|H|c_r\rangle$, folded over all pairs into one set $(\gamma^\zeta, \Gamma^\zeta)$.

The Z-vector. Right-hand sides: orbital $g^\rho + \sum \zeta_{sr} g^{(sr)} + \mathrm{pack}\, A$, g^ρ the Fock's orbital gradient at the ρ-weighted RDMs; CI $\bar{y}_r/(2w_r)$ for every root (`zvector`'s CI rows are the SA Hessian's divided by $2w_r$); rotations t. At $n = 1$ this is §6's $\frac{1}{2}(y + y^c)$. A root of weight below the rotation tolerance is refused, since it has no SA CI row. The gate is the single-state one at the ρ-weighted RDMs plus ζ's and z's.

Two derivative passes.

$$\frac{dE_K}{dR} = \underbrace{\texttt{df_gradient}\big(\rho;\ (Z, z, \Theta);\ \mathrm{extra}\big)}_{\text{reference at } \rho\text{, multipliers, } \mathcal{E}^{\mathrm{corr}}+\Phi^{\mathrm{SA}}} + \underbrace{\texttt{df_transition_gradient}\big(\gamma^\zeta, \Gamma^\zeta\big)}_{\zeta\text{, skipped when } \zeta=0}. \quad (46)$$

8.3 The W form, and several W at once

Everything in $\mathcal{L}$ is linear in W except the core energy, which $\sum_M \rho_M E_M$ carries at $\mathrm{tr}\, W$. The W form (`qd_*_gradient(..., w)`) takes any symmetric W: $\rho = \mathrm{diag}\, W$; g^ρ and the gate carry the core part $\mathrm{tr}\, W$ times; and the first pass is one `df_gradient` at the mix ρ when $\sum \rho = 0$ (its core-off path) or $\sum \rho = 1$ with $\rho \geq 0$. Any other $\rho = (\mathrm{tr}\, W)p + d$, $p \geq 0$ normalised, takes one unrelaxed pass at p scaled by $\mathrm{tr}\, W$ and one relaxed pass at d. So $G(W_1 + W_2) = G(W_1) + G(W_2)$ and $\sum_K dE_K/dR = G(1)$. The state form is a wrapper that builds $W = u_K u_K^{\mathsf{T}}$, refusing a QD state within $10^{-8}\, E_h$ of another.

A list of W's shares the forward arrays, the Fock, the CI-relaxed Hessian and one block Z-vector solve (`response::zvectors`) over all right-hand sides, each column's true residual checked against 10^{-10}. The derivative passes stay one per W: one three-centre derivative kernel is most of each pass, so stacking them would need a multi-density kernel.

The dropped rank-projector terms (PC). Eq. (17) and its QD form differentiate the restricted inverse as if S_N's kept span were fixed, exact when S_N's dropped modes annihilate $|N\rangle$, at eigenstates. Ladders along each real root c_r are what check it, since random RDM directions carry no eigenstate structure. The residual terms of an unconverged CASCI are the ones Kollmar et al. [12] show reach the 5-RDM in the Dyall matrix elements.

Algorithm 4 The DF-QD-NEVPT2 gradient $G(W)$ (`qd_lagrangians` and `qd_passes`)

Input: DF-SA-CASSCF $(C, \{c_r\}, w, \{E_r\})$, core and virtuals canonical over f^{SA}; SC or PC; symmetric W (state form: $W = u_K u_K^{\mathsf{T}}$ from H^{eff})

Output: $G(W)$; for the state form dE_K/dR

1: $(\bar{h}, \Lambda, \bar{y}_r, X, A, \varepsilon) \leftarrow$ `qd_sc_adjoint` or `qd_pc_adjoint` at W
2: $z^c \leftarrow$ Eq. (24) over A and $\varepsilon^{\mathrm{SA}}$; Φ^{SA}'s adjoints at the weights; add
3: τ; ζ on equal-weight pairs, t on unequal ones; ζ's orbital gradient and $(\gamma^\zeta, \Gamma^\zeta)$
4: $\rho \leftarrow \operatorname{diag} W$; g^ρ at the ρ-weighted RDMs, core part at $\operatorname{tr} W$
5: $(Z, z, \Theta) \leftarrow$ `zvector`$(g^\rho + \zeta \cdot g + \operatorname{pack} A,\ \bar{y}_r/(2w_r),\ t)$
6: **gate** per block
7: $G \leftarrow$ `df_gradient`$(\rho, (Z, z, \Theta), \mathrm{extra})$ + `df_transition_gradient`$(\gamma^\zeta, \Gamma^\zeta)$

9 Interstate and derivative couplings

9.1 The interstate coupling

With $W^{PQ} = \frac{1}{2}(u_P u_Q^{\mathsf{T}} + u_Q u_P^{\mathsf{T}})$ (Park's R_{PQMN}),

$$\mathbf{h}_{PQ} = u_P^{\mathsf{T}}(\nabla H^{\mathrm{eff}})u_Q = G(W^{PQ}), \qquad \operatorname{tr} W^{PQ} = 0, \tag{47}$$

so the first pass takes `df_gradient`'s core-off path: the reference part $\sum_r R_{rP} R_{rQ} \nabla E_r$ has no core energy and no nuclear repulsion. Since $u_P^{\mathsf{T}} \delta H^{\mathrm{eff}} u_Q = (E_Q - E_P) u_P^{\mathsf{T}} \delta u_Q$,

$$\mathbf{d}_{PQ}^{\mathrm{NEV}} = \sum_M R_{MP} \nabla R_{MQ} = \frac{\mathbf{h}_{PQ}}{E_Q - E_P}. \tag{48}$$

A pair within $10^{-8}\,E_h$ is refused. $\mathbf{h}_{PQ}$ carries R_P's sign times R_Q's, and so the roots' phases only through them.

9.2 The derivative coupling

With $\tilde{\Psi}_P = \sum_M (|M\rangle + |\Psi_M^{(1)}\rangle) R_{MP}$, Park's Eqs. 45–50 [10] (after the MS-CASPT2 derivation of Park and Shiozaki [11]) give

$$\mathbf{d}_{PQ} = \mathbf{d}_{PQ}^{\mathrm{NEV}} + \mathbf{d}_{PQ}^{\mathrm{CAS}} + \mathbf{d}_{PQ}^{\mathrm{PT2}}, \qquad T_{MN}^{PQ} = \tfrac{1}{2}(R_{MP} R_{NQ} - R_{MQ} R_{NP}), \tag{49}$$

$$\mathbf{d}_{PQ}^{\mathrm{CAS}} = \sum_{MN} T_{MN}^{PQ} \langle M | \nabla N \rangle, \qquad \mathbf{d}_{PQ}^{\mathrm{PT2}} = \sum_{MN} T_{MN}^{PQ} \langle \Psi_M^{(1)} | \nabla N \rangle = \sum_{xy} D_{xy}^{\mathrm{PT2}} \langle \phi_x | \nabla \phi_y \rangle, \tag{50}$$

the last because $\Psi_M^{(1)}$ lies outside the active space, so ∇ reaches it only through $|N\rangle$'s orbitals, with

$$D_{xy}^{\mathrm{PT2}} = \sum_{MN} T_{MN} \langle \Psi_M^{(1)} | E_{xy} | N \rangle. \tag{51}$$

Both $\mathbf{d}^{\mathrm{CAS}}$ and $\mathbf{d}^{\mathrm{PT2}}$ are one `mcscf::df_nac` combination

$$\mathbf{d}^T = \sum_{MN} T_{MN} \mathbf{d}_{MN} + \mathbf{d}[D^X], \qquad \mathbf{d}[D^X] = \sum_{xy} D_{xy}^X \langle \phi_x | \nabla \phi_y \rangle, \tag{52}$$

with the SA orbitals and CI relaxed: $\mathbf{d}^{\mathrm{CAS}}$ at $(T^{PQ}, D^X = 0)$ and $\mathbf{d}^{\mathrm{PT2}}$ at $(T = 0, D^X = D^{\mathrm{PT2}})$. Its pieces, by linearity in the pair coupling:

- **CI part:** the transition Lagrangian at the combined RDMs $\sum_{M<N} 2T_{MN}/(E_N - E_M)\,(\gamma^{(MN)}, \Gamma^{(MN)})$ (symmetrised), one Z-vector and one `df_transition_gradient`.
- **CSF part:** $\frac{1}{2}G[D^{\mathsf{T}} - D]$ at $D = C_a \gamma^T C_a^{\mathsf{T}}$, $\gamma^T = \sum_{MN} T_{MN}\gamma^{(MN)}$, with $G[a]_{A,k} = \sum_{m\in A}\sum_n a_{mn}\,\partial S_{mn}/\partial A_k$ the half-derivative overlap (PySCF's `nac_csf`).
- D^X**:** with $C(R) = C_0(C_0^{\mathsf{T}} S(R) C_0)^{-1/2} e^{\kappa}$, the *basis motion* $\frac{1}{2}G[C(D^{X\mathsf{T}} - D^X)C^{\mathsf{T}}]$, and the *orbital response*, $D^X_{pq} - D^X_{qp}$ added to the Z-vector's orbital right-hand side. D^X has no CI part.

`mult_ediff`	`use_etfs`	`total`
false	false	$\mathbf{h}_{PQ}/\Delta E + \mathbf{d}^{\mathrm{CAS}} + \mathbf{d}^{\mathrm{PT2}}$ (default)
false	true	$\mathbf{h}_{PQ}/\Delta E + \mathbf{d}^{\mathrm{CAS}} + \mathbf{d}^{\mathrm{PT2}} - \mathrm{csf}$
true	false	$\mathbf{h}_{PQ} + \Delta E(\mathbf{d}^{\mathrm{CAS}} + \mathbf{d}^{\mathrm{PT2}})$
true	true	$\mathbf{h}_{PQ} + \Delta E(\mathbf{d}^{\mathrm{CAS}} + \mathbf{d}^{\mathrm{PT2}} - \mathrm{csf})$

$\Delta E = E_Q - E_P$, and csf is $\mathbf{d}^{\mathrm{CAS}}$'s CSF part plus $\mathbf{d}^{\mathrm{PT2}}$'s basis motion (the terms electron-translation factors drop).

9.3 The PT2 transition density

T is antisymmetric, so only $M \neq N$ enter and every term that reads no RDM carries $\langle M|N\rangle = 0$. $E_{xy}|N\rangle$ lies in the first-order interacting space only for $(x,y) = (t,i)$, (a,t) and (a,i), one class each (Si, Sr, Sir); every other element of D^{PT2} is exactly zero. In that class $E_{xy}|N\rangle$ is the one-body function χ_y^N of V's h_v term, and $\Psi_M^{(1)}$ is a combination of the class's internal functions ϕ_μ^M (and, for SC, of χ^M). So with the T-weighted transition RDMs $\rho^M = \sum_N T_{MN}\gamma^{(MN)}$ (dm1 and dm2, bra M), each block needs only

$$\langle \phi_\mu^M | E_{xy} | \rho^M \rangle = S_1(\rho^M)[\mu, y], \qquad \langle \chi_a^M | E_{xy} | \rho^M \rangle = S_{11}(\rho^M)[a, y], \tag{53}$$

S_1 the class's PC one-body metric and S_{11} SC's $h_v \cdot h_v$ metric, both over ρ^M. No dm3, no f3, no 4-RDM.

class	block	$S_1(\rho)$	$S_{11}(\rho)$
Sr, over r	$D[r,u]$	$\rho_2[r',p,q,u]$, $\mu = (p,q,r')$	$\rho_1[a,u]$
Si, over i	$D[u,i]$	$\mathrm{dm2_h}(\rho)[r,p,q,u]$	$-\rho_1[u,a]$
Sir, over (r,i)	$D[r,i]$	$[2\rho_1[q,p];\ -\rho_1[q,p]]$, $\mu = (x,p,q)$	0

with $\mathrm{dm2_h}(\rho)[a,b,c,d] = 2\rho_1[a,b]\delta_{cd} - \rho_2[a,b,d,c]$.

SC. With the handed-over c_t^M and slab rows x,

$$D[r,u] = \sum_M c_r^M \Big(\sum_{pqr'} x[r,p,q,r']\,\rho_2^M[r',p,q,u] + \sum_a (h_v^{\mathrm{Sr}})_{ra}\,\rho_1^M[a,u] \Big), \tag{54}$$

$$D[u,i] = \sum_M c_i^M \Big(\sum_{pqr} x[q,p,i,r]\,\mathrm{dm2_h}(\rho^M)[r,p,q,u] - \sum_a (h_v^{\mathrm{Si}})_{ai}\,\rho_1^M[u,a] \Big), \tag{55}$$

$$D[r,i] = \sum_M c_{ri}^M \sum_{pq} \big(2x_1[r,p,i,q] - x_2[r,p,q,i]\big)\rho_1^M[q,p], \tag{56}$$

one more walk of the Sr, Si, Sir1 and Sir2 slabs.

PC. $\Psi_M^{(1)} \ni -\sum_k g_{tk}^M \sum_\mu U_M[\mu,k]\phi_\mu$ has no χ part, so with $W^M = S_1(\rho^M)^\mathsf{T} U_M$, $D[\cdot,t] = -\sum_M \sum_k W^M[\cdot,k]\, g_{tk}^M$: two einsums per state and class, no slab.

9.4 Three solves or one: the fold

The default path solves three Z-vectors per pair ($\mathbf{h}_{PQ}$'s, and the combination twice) so that $\mathbf{h}_{PQ}$, $\mathbf{d}^{\mathrm{CAS}}$ and $\mathbf{d}^{\mathrm{PT2}}$ can be reported apart. Every right-hand side is linear in its source and all three share the CI-relaxed SA Hessian, so Park's Eqs. 51–52 [10] fold them:

$$\mathbf{h}_{PQ} + \Delta E(\mathbf{d}^{\mathrm{CAS}} + \mathbf{d}^{\mathrm{PT2}} - \mathrm{csf}) = G(W^{PQ})\Big|_{\substack{\mathrm{RDMs}+\Delta E(\gamma,\Gamma)_{T^{PQ}} \\ A+\Delta E(D^{\mathrm{PT2}} - D^{\mathrm{PT2T}})}}, \tag{57}$$

one Z-vector whose orbital right-hand side and Hellmann–Feynman pass take ΔE times the combination's RDMs, and whose A takes ΔE times D^{PT2}'s antisymmetric part; csf needs no Z-vector. The fold's RDMs are no state's, so their own active–active asymmetry (the CSF term's, which no multiplier removes) is taken out of the gate in that block. It costs about 0.55–0.68 of the three-solve path, and is off by default because it cannot report the three parts separately.

Algorithm 5 The QD-NEVPT2 derivative coupling (`qd_sc_nac`, `qd_pc_nac`)

Input: DF-SA-CASSCF; SC or PC; QD states $P \neq Q$; `NacOptions`; fold
Output: $\mathbf{d}_{PQ}$ and its parts

H^{eff}, R (signed), $\Delta E = E_Q - E_P$; refuse $|\Delta E| < 10^{-8}$
$T^{PQ} \leftarrow \frac{1}{2}(R_P R_Q^\mathsf{T} - R_Q R_P^\mathsf{T})$ ▷ exactly antisymmetric
$D^{\mathrm{PT2}} \leftarrow$ Eq. (51) at T^{PQ} over the handed-over amplitudes
if fold **then**
    $(\gamma, \Gamma, \mathrm{csf}) \leftarrow$ `nac_combination_terms`$(T^{PQ}, D^{\mathrm{PT2}})$
    $F \leftarrow G(W^{PQ})$ with $\Delta E(\gamma,\Gamma)$ and $\Delta E(D^{\mathrm{PT2}} - D^{\mathrm{PT2T}})$ folded in
    `total` $\leftarrow F/\Delta E + \mathrm{csf}$ (per the options)
else
    $\mathbf{h}_{PQ} \leftarrow G(W^{PQ})$
    $\mathbf{d}^{\mathrm{CAS}} \leftarrow$ `df_nac`(T^{PQ}); $\mathbf{d}^{\mathrm{PT2}} \leftarrow$ `df_nac`$(0, D^{\mathrm{PT2}})$
    `total` $\leftarrow \mathbf{h}_{PQ}/\Delta E + \mathbf{d}^{\mathrm{CAS}} + \mathbf{d}^{\mathrm{PT2}}$ (per the options)

Conventions. Bra P, ket Q: $\mathbf{d}_{PQ} = -\mathbf{d}_{QP}$, $\mathbf{h}_{PQ} = \mathbf{h}_{QP}$ (PySCF's `kernel(state=(I,J))` is $\langle I|\partial J\rangle$). $\mathbf{d}_{PQ}$ changes sign with the CI phases as R carries them, by $s_{m(P)}s_{m(Q)}$, $m(P)$ the index of R_P's largest component; no phase is fixed by the call.

10 Conical intersections

`qd_*_intersection` returns $\mathbf{g}_P = G(u_P u_P^\mathsf{T})$, $\mathbf{g}_Q = G(u_Q u_Q^\mathsf{T})$ and $\mathbf{h}_{PQ} = G(W^{PQ})$ from one set of H^{eff} eigenvectors, by default through one several-W call (one shared block Z-vector). It does not refuse a degenerate pair: G is linear in W and nothing divides by $E_Q - E_P$. At degeneracy u_P, u_Q are any orthonormal pair of the two-state space and the three vectors turn with them, but the branching plane $\mathrm{span}\{\mathbf{g}_Q - \mathbf{g}_P, \mathbf{h}_{PQ}\}$ and $\frac{1}{2}(\mathbf{g}_P + \mathbf{g}_Q)$ do not, which is all `geometry::optimize_meci` uses.

Gap sensitivities. An element $a_W = \mathrm{tr}(WH^{\mathrm{eff}})$ is not stationary in the SA-CASSCF parameters, so a solution with residual $\mathcal{G}$ moves it by $\delta a_W = -z^W \cdot \mathcal{G}$ to first order, z^W the W-form's Z-vector. In the pair's 2×2 block $[[a, b], [b, d]]$ the gap is $\sqrt{(a-d)^2 + 4b^2}$, and

$$|\delta\,\mathrm{gap}| \le s_o\|g\| + s_c\sqrt{n}\,\rho_{\mathrm{CI}}, \qquad s_o = \sqrt{\|z_o^Q - z_o^P\|^2 + 4\|z_o^{PQ}\|^2}, \quad s_c = \sqrt{\|\hat{z}_c^Q - \hat{z}_c^P\|^2 + 4\|\hat{z}_c^{PQ}\|^2}, \tag{58}$$

$\hat{z}_{c,r} = 2w_r z_{c,r}$, $\|g\|$ the SA orbital gradient norm and ρ_{CI} the largest Davidson residual. Θ needs no term (Rayleigh–Ritz) and z^c none (a direct diagonalisation). Both s are invariant to turning u_P, u_Q within their pair.

Inside a degenerate CASSCF pair (#902). A QD pair can be healthy while the CASSCF pair under it is not: at ethylene's C_{3v} MECI the QD gap is 6.6×10^{-10} but the CASSCF ^{1}E pair is $7.9 \times 10^{-9}\,E_h$ apart, below `mcscf::kRootGapFloor` (10^{-8}), where the root-pair multipliers are refused. Write $E_K(R) = \tilde{E}_K(R, \theta(R))$, θ the references' rotation within the pair. $\tilde{E}_K$ is smooth, but at the reference seam θ turns by half the polar angle in the reference branching plane, so $dE_K/dR = \partial_R\tilde{E}_K + \tau_K\partial_R\theta$ with $\partial_R\theta \sim 1/\rho$. SC's g_K diverge; PC's are bounded, but their limit depends on the direction of approach. Only PC's $\tau_P + \tau_Q$ vanishes (measured at $\le 5 \times 10^{-12}$), so only $\bar{\mathbf{g}} = \frac{1}{2}(\mathbf{g}_P + \mathbf{g}_Q)$ has a limit, and $\bar{\mathbf{g}}$ is all a MECI walk's convergence test reads. So when the closest CASSCF pair is under the floor and carries more than half of u_P's and u_Q's weight, `qd_pc_intersection` makes one W-form call at $W = \frac{1}{2}(u_P u_P^{\mathsf{T}} + u_Q u_Q^{\mathsf{T}})$ with that pair's ζ set to zero and its τ gated at 10^{-11} (`kLimitTauGate`), and returns `g_mean` and `reference_pair_tau` with $\mathbf{g}_P$, $\mathbf{g}_Q$, $\mathbf{h}_{PQ}$ empty and s_o, s_c NaN. `optimize_meci` carries the branching plane from its last evaluation, re-orthonormalised against the rigid motions, and drops the gap from its trust ratio at such an evaluation. SC, and a degenerate pair that is not the QD pair's, keep the refusal. The #847 walk that stopped at evaluation 10 now converges there ($\tau = 5.6 \times 10^{-18}$, QD gap 4.8×10^{-11}), 3.1×10^{-5} Å from exact C_{3v}.

11 The QD-NEVPT2 nuclear Hessian

The reference is a DF-SA-CASSCF over n roots r, s, M, N with weights w_r, CI vectors c_r and energies E_r. The couplings C_{MN} are Eqs. (7)–(8), H^{eff} is Eq. (9), and its eigenpairs $H^{\mathrm{eff}}R_P = E_P R_P$ are the QD states P, Q (R_P is §8's u_P). The functional is $\mathcal{E}[W]$ of Eq. (43), $E_P = \mathcal{E}[W^P]$ with $W^P = R_P R_P^{\mathsf{T}}$, and its Lagrangian $\mathcal{L}[W]$ is Eq. (44), with multipliers $\lambda[W] = (Z, \{z_r\}, \Theta, \zeta, z^c)$ and, at unequal weights, the rotation unknowns t inside Z's solve. In this section and the next two N_{at} counts atoms, because N is a root; A, B are nuclear coordinates.

What is built. `hessian::df_qd_sc_nevpt2` (#965) is the analytic Hessian of an SC-QD-NEVPT2 state, all-electron, in Cartesian or caller-given coordinates, assembled in `hessian` as the single-state Hessians are (§7.4) from: the SA parameter response (`df_sacasscf_cp`, `df_qd_nevpt2_tangents`, #961); the builders' forward tangent over a state average (`qd_builders_tangent`, #958); the pair walk's forward tangents (`sc_effective_hamiltonian_tangent`, #954; `pc_effective_hamiltonian_tangent`, #960) and Hessian-vector products (`sc_effective_hamiltonian_hvp`, #962; `qd_sc_hessian_product`, #963; `pc_effective_hamiltonian_hvp` and `qd_pc_hessian_product`, #964); the multipliers' CASSCF terms (`df_sacasscf_z` at a QD Lagrangian, #956); the ζ term (`df_qd_zeta`, #955); and term (ii) (`df_qd_eigenvector_response`, #957). Not yet in the tree: the PC-QD Hessian `df_qd_pc_nevpt2` (#966), every piece of which above is; the pair-mean Hessian and the seam's 2×2

at a crossing (Eqs. (71) and (72), #967); and the analytic `[hessian]` over a `quasi_degenerate [nevpt2_gradient]` (#968). The pipeline's semi-numerical QD Hessian is in the tree (#959, §13).

11.1 The eigenvalue's second derivative

Write $\mathsf{H}^A = dH^{\text{eff}}/dR_A$ and $\mathsf{H}^{AB} = d^2H^{\text{eff}}/dR_A\,dR_B$. These are *total* derivatives, element by element, with every SA-CASSCF parameter relaxed. Write $\tilde{\mathsf{h}}^A = R^{\mathsf{T}}\mathsf{H}^A R$ for the first derivative in the QD eigenbasis. Its elements are the interstate couplings, $\tilde{\mathsf{h}}^A_{PQ} = \mathbf{h}_{PQ,A}$ (Eq. (47), $\mathcal{E}[W^{PQ}]$'s gradient), and $\tilde{\mathsf{h}}^A_{PP} = dE_P/dR_A$.

For a simple eigenvalue E_P, in the gauge $R_P^{\mathsf{T}}\,dR_P = 0$,

$$\frac{dR_P}{dR_A} = \sum_{Q\neq P} R_Q\,\frac{\tilde{\mathsf{h}}^A_{QP}}{E_P - E_Q}, \tag{59}$$

and differentiating $dE_P/dR_A = R_P^{\mathsf{T}}\mathsf{H}^A R_P$ once more gives

$$\boxed{\frac{d^2E_P}{dR_A\,dR_B} = \underbrace{R_P^{\mathsf{T}}\,\mathsf{H}^{AB}R_P}_{\text{(i): } d^2\mathcal{E}[W^P] \text{ at fixed } W} + \underbrace{2\sum_{Q\neq P}\frac{\tilde{\mathsf{h}}^A_{PQ}\,\tilde{\mathsf{h}}^B_{PQ}}{E_P - E_Q}}_{\text{(ii): the QD eigenvector response}}}\,. \tag{60}$$

The expression is symmetric in A and B, and term (ii) has the sign of second-order perturbation theory: the lower state's curvature goes down and the upper state's goes up.

Term (ii) needs no new machinery. It is built from first derivatives only, the $n-1$ couplings $\mathbf{h}_{PQ}$ that `qd_sc_coupling` and `qd_pc_coupling` return ($G(W^{PQ})$, trace-zero W, no core energy). Over all P that is the $n(n-1)/2$ off-diagonal elements of $\tilde{\mathsf{h}}$. Because $\mathcal{E}$ is linear in W, `df_qd_eigenvector_response` (#957) takes them all through one call of the several-W form (§8), one shared Z-vector solve, each W^{PQ}'s single core-off `df_gradient` pass: the list form takes trace-zero Ws as they stand. It returns term (ii) and Eq. (59) for every state, the couplings, and the sum rule below as a diagnostic. R, like W, needs no multiplier and no response solve: the $n \times n$ eigenproblem is differentiated in closed form, on the host. A pair of QD states closer than `mcscf::kRootGapFloor` ($10^{-8}\,E_h$) is refused, naming both, before any adjoint is built.

Term (i) is Eq. (34) at fixed W. That derivation used only $\mathcal{L}_x = 0$ and $g = 0$ at the solution. Both hold for $\mathcal{L}[W]$ at any symmetric W, since the gradient's multipliers make $\mathcal{L}[W]$ stationary in every SA-CASSCF parameter, including the rotations among the roots. So

$$R_P^{\mathsf{T}}\mathsf{H}^{AB}R_P = \mathcal{L}_{AB} + \mathcal{L}_{Ax}\,x_B + x_A^{\mathsf{T}}\mathcal{L}_{xB} + x_A^{\mathsf{T}}\mathcal{L}_{xx}\,x_B, \qquad \mathcal{L} = \mathcal{L}[W^P], \quad g_x\,x_A = -g_A, \tag{61}$$

where x and g are now the SA-CASSCF's (§11.2).

Two linearities set the cost.

- **x_A does not depend on W.** The responses are the reference's, so one block CP-SA-CASSCF solve over $3N_{\text{at}}$ right-hand sides serves every QD state.

- $\lambda[W]$ **is linear in** W**.** The multiplier equations do not depend on W, and their sources (A^{corr}, $\bar{y}_r$, τ, g^ρ) are linear in it. So $\mathcal{L}[W]$, and with it every term of Eq. (61), is linear in W. Term (i) for all n states is $\sum_{MN} R_{MP} R_{NP}\, \mathsf{H}^{AB}_{MN}$ over the $n(n+1)/2$ element Hessians, or Eq. (61) once per wanted P. Each costs about the same, and there are at most $n \le n(n+1)/2$ states, so per-state is never worse, and it is what `df_qd_sc_nevpt2` does. Its W overload is term (i) at any symmetric W whose diagonal is a mixture, the form Eq. (72)'s seam 2×2 needs.

The sum rule. Term (ii) is antisymmetric under $P \leftrightarrow Q$ term by term, so it cancels in the sum over states:

$$\sum_P \frac{d^2 E_P}{dR_A\, dR_B} = \frac{d^2\, \mathcal{E}[1]}{dR_A\, dR_B} = \frac{d^2 \operatorname{tr} H^{\mathrm{eff}}}{dR_A\, dR_B}. \tag{62}$$

This is the second-order form of the gradient's $\sum_K dE_K/dR = G(1)$ (§8), and it checks term (ii) against term (i) with no oracle. At a crossing the individual E_P have no Hessian, but $\operatorname{tr} H^{\mathrm{eff}}$ and the pair's sum do (Eq. (71)), so this is the check that still runs there (§10's ethylene MECI).

11.2 The parameter response on a state average

The parameters are $x = (\kappa_{\mathrm{nr}}, \kappa_{oo}, \{c_r\})$. A CI direction is taken jointly over the roots, in the exponential parametrisation of the root set,

$$c_r(K) = e^K c_r, \qquad K = \sum_r (\xi_r c_r^{\mathsf{T}} - c_r \xi_r^{\mathsf{T}}) + \sum_{r<s} \theta_{sr} (c_s c_r^{\mathsf{T}} - c_r c_s^{\mathsf{T}}), \qquad \xi_r \perp \text{every root}. \tag{63}$$

So ξ_r is root r's out-of-span move, and θ_{sr} mixes c_s into c_r. `df_qd_nevpt2_tangents` (#961) forms the response x_A and the total tangent t_A of every input of H^{eff}, $(h, B, \{c_r\})$, as Eq. (36) with $\dot{c}_r^A = \xi_r^A + \sum_s \theta_{sr}^A c_s$. The response has three parts, block triangular as at $n = 1$ (§7.2):

1. **CP-SA-CASSCF**, $3N_{\mathrm{at}}$ right-hand sides, over $(\kappa_{\mathrm{nr}}, \{\xi_r\})$ and the θ_{sr} of pairs with $w_r \neq w_s$, which the SA orbital Hessian carries as its Θ rows (`response::zvector`'s convention). This is `df_sacasscf_cp`, the response `hessian::df_sacasscf` forms for one root of a state average, stopped there. It takes weights equal for some pairs only, such as $(0.4, 0.4, 0.2)$: each pair is rotating or not by `response::rotating_pairs`, and an equal pair's row is left out of the solve. It does not depend on the state, so one solve serves every QD state.

2. **Equal-weight rotations**, closed form once 1. is done. E_{SA} is blind to them, so the condition $\langle c_s | H | c_r \rangle = 0$ fixes them. Along Eq. (63) that element moves by $\theta_{sr}(E_s - E_r)$, hence

$$\theta_{sr}^A = \frac{\mathcal{H}_{sr}^A}{E_r - E_s} \qquad (w_r = w_s), \tag{64}$$

   where $\mathcal{H}_{sr}^A$ is the total first-order change of $\langle c_s | H | c_r \rangle$ at $\theta = 0$: the derivative integrals, the connection and κ_{nr}^A, through the active Hamiltonian's tangent (`qd_builders_tangent`'s one- and two-electron parts) against the transition RDMs. The out-of-span moves contribute nothing: $\langle \xi_s^A | H | c_r \rangle = E_r \langle \xi_s^A | c_r \rangle = 0$, because $\xi \perp$ every root. That holds only at eigenvectors, so the reference must be converged tightly (the tests take $\|g\| < 10^{-10}$). An unequal pair's Θ meets the same condition: its CP row is $(E_s - E_r)\Theta_{sr} + c_s \cdot H' c_r + \tilde{H}_{sr}^A = 0$.

3. **Pseudocanonical at the SA Fock**, closed form once 1. and 2. are done: Eq. (35) with $f \to f^{\mathrm{SA}}$ and $\varepsilon_p = f_{pp}^{\mathrm{SA}}$, the solution canonical in f^{SA}. The response of $\gamma^{\mathrm{SA}} = \sum_r w_r \Gamma_r^{(1)}$ along x_A is

$$\dot{\gamma}_A^{\mathrm{SA}} = \sum_r w_r\, \Gamma^{(1)}(\xi_r^A, c_r)^{\mathrm{sym}} + \sum_{r<s} (w_r - w_s)\, \theta_{sr}^A\, 2\gamma^{(sr)}, \tag{65}$$

with $\Gamma^{(1)}(u,c)^{\mathrm{sym}} = \langle u|E|c\rangle + \langle c|E|u\rangle$ the tangent RDM. So f^{SA} sees an equal-weight rotation not at all, and an unequal one only through $w_r - w_s$; the tangent of f^{SA} is taken before step 2's θ is added and serves both. The degenerate-pair rule is the single-state one ($10^{-6}\,E_h$, 10^{-8}): SC refuses, naming the pair and the coordinate, and PC records it.

Eq. (64) is a second 1/gap, this time the CASSCF root gap, and it enters Eq. (61) quadratically. Unlike in a CASSCF Hessian it is not suppressed by the stationarity of E_r: $\mathcal{E}^{\mathrm{corr}}[W]$ reads each c_r individually, which is the reason the gradient carries ζ at all. Two roots under `mcscf::kRootGapFloor` apart are refused before anything runs, and a rotation $|\theta^A_{sr}|$ above 10 per bohr, the CP solve's or Eq. (64)'s, is refused, naming the pair and the coordinate (`root_rotation`'s bound, where the quadratic model holds within about 1/20 bohr). `theta` returns every rotation for the $\mathcal{L}_{xx}$ actions, and `max_theta`, `max_kappa_oo` are reported.

11.3 The parameter Hessian $\mathcal{L}_{xx}[W]$

$$\mathcal{L}_{xx}[W] = \underbrace{\frac{\partial^2 \mathcal{E}[W]}{\partial x\,\partial x}}_{\text{level 1, over pairs}} + \underbrace{Z\cdot\frac{\partial^2 g_{\mathrm{SA}}}{\partial x\,\partial x} + \sum_r z_r\cdot\frac{\partial^2 g_{\mathrm{ci},r}}{\partial x\,\partial x}}_{\text{third derivative of } E_{\mathrm{SA}}} + \underbrace{\sum_{\substack{r<s\\ w_r=w_s}} \zeta_{sr}\,\frac{\partial^2\langle c_s|H|c_r\rangle}{\partial x\,\partial x}}_{\text{a CASSCF Hessian at transition RDMs}} + z^c\cdot\frac{\partial^2 f^{\mathrm{SA}}}{\partial x\,\partial x}. \tag{66}$$

As at $n = 1$ only the actions $x_A^{\mathsf{T}}\mathcal{L}_{xx}x_B$ are formed.

The CI curvature on the root set. The single-state sphere's $-(u\cdot v)\,c$ (§7.3) generalises along Eq. (63) to

$$\frac{\partial^2 c_r}{\partial K_u\,\partial K_v} = \tfrac{1}{2}\{K_u, K_v\}\,c_r, \tag{67}$$

contracted with the raw $\partial\mathcal{L}/\partial c_r$ of every root. At $n = 1$, $\frac{1}{2}\{K_u, K_v\}c = -(u\cdot v)\,c$, the sphere's term. Along two rotations it is not zero. For example, θ_{sr} against itself gives $-\theta^2(c_r\,\partial_{c_r} + c_s\,\partial_{c_s})\mathcal{L}$. The off-shell extension is again the gradient's own (raw $\bar{y}_r$, the rank-reduced one; §7.3, "Φ, and the functional of c"), so a Hessian built on these products and the gradient's multipliers differentiates one functional. `qd_sc_hessian_product` takes a raw $\dot{c}$ to the generators' $u_r = \dot{c}_r - \frac{1}{2}\sum_s c_s(c_s\cdot\dot{c}_r + c_r\cdot\dot{c}_s)$ (an orthogonal projection, $K(\dot{c}) = K(u)$), and returns the gradient over raw CI directions of the Hessian in K, the curvature its last part. `mcscf::ThirdDerivative`'s joint exponential already is Eq. (67).

The RDMs. A transition RDM $\Gamma^{(MN)}$ is bilinear in (c_M, c_N), so along directions $u = \{u_r\}$ and $v = \{v_r\}$

$$\frac{\partial^2\Gamma^{(MN)}}{\partial u\,\partial v} = \Gamma(u_M, v_N) + \Gamma(v_M, u_N), \qquad \frac{\partial^3\Gamma^{(MN)}}{\partial u\,\partial v\,\partial w} = 0, \tag{68}$$

with $\Gamma(a,b)$ `nevpt2::rdms` at bra a, ket b. The single-state "no tower" property survives: everything is quadratic in the roots, the digests are linear in the integrals, and ε is affine in γ^{SA}. What is new is the off-diagonal finishes. In the pair walk (§4.4, §5.4) each unordered pair's 2×2 block takes the tangent builders at two directions on its transition RDMs.

The SC off-diagonal finish. Per perturber t of ket N and bra M, the coupling of Eq. (7) is $g = -T\,N/d$ with $T = \bar{T}^t_{MN}$, $N = N^N_t$, $H = H^N_t$, $\Delta = \Delta_t$ and $d = N\Delta + H$, differentiated in that

form, with no $1/N$. Its first derivatives, §5.4's SC-QD multipliers in this form, are

$$g_T = -\frac{N}{d}, \qquad g_N = -\frac{TH}{d^2}, \qquad g_H = \frac{TN}{d^2}, \qquad g_\Delta = \frac{TN^2}{d^2},$$

and its second derivatives are

$$\begin{aligned} &g_{TT} = 0, && g_{TN} = -\frac{H}{d^2}, && g_{TH} = \frac{N}{d^2}, && g_{T\Delta} = \frac{N^2}{d^2}, \\ &g_{NN} = \frac{2TH\Delta}{d^3}, && g_{NH} = \frac{T(H - N\Delta)}{d^3}, && g_{N\Delta} = \frac{2TNH}{d^3}, \\ &g_{HH} = -\frac{2TN}{d^3}, && g_{H\Delta} = -\frac{2TN^2}{d^3}, && g_{\Delta\Delta} = -\frac{2TN^3}{d^3}. \end{aligned} \tag{69}$$

Each is weighted by W_{MN}. The T bars split $\frac{1}{2}$, $\frac{1}{2}$ onto the norm einsums of (M, N) and (N, M), and the ket's own (N, H, Δ) collect from every bra, its own coupling included: the diagonal energy is the same g at $T = N$, so one finish serves every coupling. `sc_effective_hamiltonian_tangent` (#954) gives the first-order $(\dot{T}, \dot{N}, \dot{H}, \dot{\Delta})$ along each direction, and `sc_effective_hamiltonian_hvp` (#962) takes the multipliers' tangent, Eq. (69) times it, onto every bar by one reverse pass at fixed multipliers, the bars at fixed multipliers by the exact line difference of §7.3; at $n = 1$ the two are `class_energies_tangent` and `class_energies_hvp`. A one-function class is $C = Tc$, $c = -1/(\Delta + \eta^N)$, with $\dot{c} = c^2(\dot{\Delta} + \dot{\eta}^N)$. Three structural facts are identities of these formulas, not separately gated (the HVP's ladders are):

- **Euler.** g is homogeneous of degree 1 in (T, N, H), so $Tg_{TX} + Ng_{NX} + Hg_{HX} = 0$ for $X \in \{T, N, H\}$: three exact zeros, where the diagonal finish (Eq. (37)) had two.
- **Rank one.** g reads (H, Δ) only through d, so that block is $g_{dd}\,(1, N) \otimes (1, N)$ with $g_{dd} = -2TN/d^3$.
- **The diagonal.** At $M = N$, where $T = N$, the total second derivatives of $g(N, N, H, \Delta)$ are Eq. (37). For example, $E_{NN} = 2g_{TN} + g_{NN}$ and $E_{N\Delta} = g_{T\Delta} + g_{N\Delta}$.

Empty tuples, on every root. A tuple ket N's guard drops ($|N_t^N| \le 10^{-14}$) that a direction u populates, by the single-state rule over ket N's tables (§7.5), takes K6 in every coupling with ket N, whatever W weighs it: $\bar{T} = -1/(\Delta + \eta)$, $\eta = H^{uu}/N^{uu}$. On the diagonal that is the single-state $\bar{P}$, $\bar{Q}$. In a coupling (M, N) it is $\bar{T}$ on T, and where the bra's tuple is empty too, T, N and H are all of second order along u, the coupling is the quadratic form $\bar{T}\,T^{uu}$, and its N and H multipliers have limits ($-\rho\eta\bar{T}^2$ and $\rho\bar{T}^2$, $\rho = T^{uu}/N^{uu}$). Where the bra's tuple is populated, T is of first order (Eq. (73)), and only $\bar{T}$ on T is formed. The refusal that goes with it is §12.4's.

The PC off-diagonal finish. Per class and tuple t, $C_{MN} \ni -\nu\,\beta$ with $\beta = b'^{\mathsf{T}} A^+ b$, $b' = b_t^{M \leftarrow N}$, $b = b_t^N$ and $A = A_{N,t} = K_N + \Delta_t S_N$ (Eq. (8)). With $x = A^+ b$ and $u = A^+ b'$, and first-order residuals $r_i = \partial_i b - \partial_i A\,x$ and $r'_i = \partial_i b' - \partial_i A\,u$ along directions $i = 1, 2$,

$$\partial_{12}\beta = x^{\mathsf{T}}\partial_{12}b' + u^{\mathsf{T}}\partial_{12}b - u^{\mathsf{T}}\partial_{12}A\,x + r_1'^{\mathsf{T}}A^+ r_2 + r_2'^{\mathsf{T}}A^+ r_1 + (\text{rank-projector terms of } A^+). \tag{70}$$

At $M = N$, where $b' = b$, $u = x$ and $r' = r$, this is the single-state PC second derivative: `pc_class_energies_hvp` is `pc_effective_hamiltonian_hvp`'s $n = 1$, $\bar{H} = 1$ (#964), and `pc_effective_hamiltonian_tangent` (#960) is the first order, $\dot{\beta} = x^{\mathsf{T}}\dot{b}' + u^{\mathsf{T}}\dot{b} - u^{\mathsf{T}}\dot{A}\,x$ plus the kept span's rotation. The three kinds of term are carried as at $n = 1$ (§7.3): the $r'^{\mathsf{T}}A^+ r$ cross terms

through the multipliers' tangent in each ket's own factorisation, $\dot{g}_M = (\dot{y}_{M\leftarrow N} - \dot{D}_N g_M)/(\lambda+\Delta)$; the ∂_{12} terms at fixed multipliers by the line difference; and the rank-projector terms through each ket's span bar, which collects every bra's, and its Daleckii–Krein derivative. They read b''s component in S_N's dropped span. That is zero at an exact null mode, but its first derivative is not where the mode lifts, which is the off-diagonal failure of §12.4. As at $n = 1$, S_N does not move along an orbital direction, so the projector terms appear in the CI–CI and CI–orbital blocks and never in the orbital–orbital one, and both S_N and the transition metrics $S^{(N,M)}$, $S_1^{(N,M)}$ move along both c_M and c_N through Eq. (68). `qd_pc_hessian_product` is `qd_sc_hessian_product` with this pair walk in place of SC's and nothing else; it refuses a ket whose kept metric spectrum reaches below 10^{-10} of its largest (#917), naming the state and the class.

The ζ term is a Hessian, not a third derivative. $\langle c_s|H|c_r\rangle$ is energy-like. Its second derivative is the CASSCF Hessian operator at the transition RDMs $(\gamma^{(sr)}, \Gamma^{(sr)})$: the orbital–orbital part is `response::Hessian`'s at those densities less its value at zero RDMs, and the CI legs are transition sigma vectors. By linearity all pairs fold into one apply at $(\gamma^\zeta, \Gamma^\zeta)$, as the gradient folds them. The transition energy is *not* stationary (its orbital gradient $g^{(sr)}$ is the gradient's ζ source), so the operator must be the full second derivative along e^κ, gradient terms included, and not a form simplified at a stationary point. `response::Hessian` is that: it is `gen_g_hop`'s operator, whose $-\frac{1}{2}(F + F^\mathsf{T})\kappa$ term carries the exponential's $\frac{1}{2}\kappa^2$ against the gradient. `df_qd_zeta` (#955) returns the term's three shares (xx, Ax, AB) at given ζ; its CI part is the second derivative of $\frac{1}{2}\sum \mathcal{Z}_{pq}\langle e^K c_q|H|e^K c_p\rangle$ ($\mathcal{Z}$ the symmetric matrix of the ζ), Eq. (67) included, from sigma vectors and two Gram matrices. Its test differences $\langle c_s|H|c_r\rangle$ itself, never the Hessian operator: differencing a Hessian-vector product applies the direction in the displaced point's frame and is not this term off stationarity. Along orbital directions alone the gradient term a stationary form would drop is 1.05×10^{-4} there, against the ladder's 10^{-7} bar.

The third derivative and z^c. $Z \cdot \partial^2 g_{\mathrm{SA}}$ and $z_r \cdot \partial^2 g_{\mathrm{ci},r}$ are the single-state level-2 term (`df_casscf_z`'s, §7.4) at the SA weights, with the curvature of Eq. (67) in place of the sphere's. They are `hessian::df_sacasscf_z`'s pieces, which since #956 take the right-hand side's weights rather than a root: one-hot on root I for `df_sacasscf`, $\rho = \operatorname{diag} W$ for a QD Lagrangian, or none for the multipliers' terms alone, at $\lambda[W^P]$'s (Z, z, Θ). At all-equal weights z carries every rotation, which makes them parameters (`ThirdDerivative::create`'s `equal_rotations`). $z^c \cdot \partial^2 f^{\mathrm{SA}}$ is Φ^{SA}'s share of the product, closed form at the weights, with $\Gamma^{(1)}$'s second derivative $\sum_r w_r[\Gamma^{(1)}(u_r, v_r) + \Gamma^{(1)}(v_r, u_r)]$, the rotations' share of it included.

11.4 The mixed and skeleton terms

$\mathcal{L}_{Ax}[W]$. The single-state parts at W, plus one:

- **NEVPT2:** the pair reverse pass (`qd_sc_adjoint`, `qd_pc_adjoint`) run on dotted integrals (h^A, B^A), forward over reverse, giving $(\dot{A}_A^{\mathrm{corr}}, \{\dot{\bar{y}}_{r,A}\})$ and the rotation source's tangent $\dot{\tau}_A$ (Eq. (45)) from them. As at $n = 1$ it is not formed on its own: each coordinate's total tangent t_A carries I^A, so $\langle t_A, \mathrm{HVP}\, t_B\rangle$ holds it (Eq. (39));
- **CASSCF:** $\partial g_{\mathrm{SA}}/\partial R_A$, the right-hand side of §11.2 step 1, already formed;
- **constraints:** Z, z_r and z^c against $\partial^2/\partial R\,\partial x$ of their conditions, as at $n = 1$ (`df_sacasscf_z`'s $I^x z$);

- **new:** $\zeta \cdot \partial^2 \langle c_s|H|c_r\rangle / \partial R_A\, \partial x$, the orbital and CI gradients of the transition energy at derivative integrals: `response::fock` at $(\gamma^\zeta, \Gamma^\zeta)$ on (h^A, B^A) plus the connection, less its zero-RDM value (`df_qd_zeta`'s `lxlambda`).

$\mathcal{L}_{AB}[W]$**.** The skeleton of Eq. (39), `df_adjoint_xy` at the total adjoint of $\mathcal{E}^{\mathrm{corr}}[W] + \Phi^{\mathrm{SA}}$, plus the reference's at ρ and the multipliers' (E_{xy} at the ρ-weighted RDMs, `df_sacasscf_z`'s L_{xy}), plus ζ's transition term at $(\gamma^\zeta, \Gamma^\zeta)$, which has no core energy (`df_qd_zeta`'s `lxy`), against the second-derivative integrals. The integrals' curvature of $\mathcal{E}^{\mathrm{corr}}[W]$ is the same forward-over-reverse pass as $\mathcal{L}_{Ax}$'s NEVPT2 part, paired with (h^B, B^B), and is again inside the product along t.

12 Where the QD Hessian exists, and where it is tractable

Each region below gets three questions:

- Is E_P twice differentiable there?
- If not, is that physics or an artefact of the method?
- Is the analytic route numerically tractable, and how does it compare with the semi-numerical one?

The single-state refusals (§7.5) apply to term (i) root by root. This section adds what QD and the state average bring.

12.1 The divisions, and how hard each one bites

Every hazard is a small number in a denominator. The Hessian sees each one at one or two powers more than the gradient does.

The near-intruder row is not hypothetical. At ethylene S_0, state 2's Sr class has its smallest $|\lambda+\Delta|$ at 9.6×10^{-3} in aug-cc-pVDZ and at 1.9×10^{-3} in aug-cc-pVTZ (`src/nevpt2/energy/README.md`, the negative-denominator table). At $1/d^3$ the second of those amplifies the finish's error by about 10^8, against 10^5 for the gradient. Higher roots in diffuse bases are the first place a QD Hessian stops being tractable even where it exists.

12.2 The QD gap: physical, and where the analytic route wins

Near a crossing, not at it. E_P is smooth, but term (ii) grows as $|\mathbf{h}_{PQ}|^2/(E_P - E_Q)$. The analytic term is closed form, so its relative error is that of $\mathbf{h}_{PQ}$, which is set by the Z-vector residual, whatever the gap. A central difference of the gradient needs a step small against the length over which the curvature changes, about $|E_P - E_Q|/|\mathbf{g}_P - \mathbf{g}_Q|$. It also needs the states tracked across the step. So the semi-numerical route fails first as the gap closes. This is the one region where the analytic Hessian buys more than exactness: it is the only usable one. *Usable* is not *meaningful*, though. Where term (ii) dominates, the surface is strongly anharmonic (a pseudo-Jahn–Teller double well), and harmonic frequencies do not describe the motion. Term (ii)'s share ($\max|\text{term (ii)}|/\max|H|$) is returned for that reason. There, the quantities worth reporting are the vibronic-model ones (the diabatic second derivatives below, and $\mathbf{h}_{PQ}$), not frequencies.

small quantity	gradient	Hessian	nature
QD gap $E_P - E_Q$	none	$1/\text{gap}$ (term (ii))	physical: a seam or avoided crossing of the corrected states
in-ensemble reference gap $E_r - E_s$	$1/\text{gap}$ (ζ, t)	$1/\text{gap}^2$ (θ^A twice)	artefact: QD-NEVPT2 is not invariant to rotations of the model space
ensemble edge $E_{n+1} - E_n$ (first excluded root)	$1/\text{gap}$ (the solve)	$1/\text{gap}^2$	artefact: the averaged ensemble changes
smallest SA-CASSCF Hessian eigenvalue μ	$1/\mu$	$1/\mu^2$	artefact: active-space or solution instability
inactive gap $\varepsilon_p - \varepsilon_q$	$1/\text{gap}$ (SC's z^c)	$1/\text{gap}^2$ (κ^A_{oo} twice)	artefact: the pseudocanonical gauge (the tree's PC too, §7.2)
perturber denominator $d_t = N_t\Delta_t + H_t$	$1/d^2$	$1/d^3$ (Eq. (69))	near-intruder: perturbation theory itself failing
PC metric eigenvalue σ near the cut	smooth	not C^2 (the lift)	artefact: the cut
symmetry-empty perturber or null mode	C^1	not bilinear	artefact: internal contraction, §12.4

Table 1: What divides, at which power. "Artefact" means a property of the method's functional, not of the Born–Oppenheimer surfaces it approximates. The analytic Hessian reproduces an artefact exactly; it does not remove one. `df_qd_sc_nevpt2` returns the QD, root, edge and inactive gaps, the smallest $|\Delta_t + H_t/N_t|$ over every state's kept perturbers, and $\max|\theta|$, $\max|\kappa_{oo}|$, $\max|\zeta|$.

At a crossing. E_P has no Hessian: it is a cone. Two related objects still exist, provided every other state stays gapped. Both are #967, not yet in the tree.

- **The pair's mean.** $\Pi = W^P + W^Q$ is the projector onto the degenerate eigenspace, well defined where R_P and R_Q separately are not. $E_P + E_Q = \mathcal{E}[\Pi]$ is smooth, and

$$\frac{d^2(E_P + E_Q)}{dR_A\, dR_B} = \Pi\text{-term (i)} + 2 \sum_{X\in\{P,Q\}} \sum_{S\notin\{P,Q\}} \frac{\tilde{\mathsf{h}}^A_{XS}\tilde{\mathsf{h}}^B_{XS}}{E_X - E_S}. \tag{71}$$

  The P–Q term of Eq. (60) cancels, as in Eq. (62), and the double sum does not depend on the basis chosen in the eigenspace.

- **The second-order effective 2×2.** In any orthonormal basis X, Y of the eigenspace, at energy E_0,

$$\frac{\partial^2 \tilde{H}_{XY}}{\partial R_A\, \partial R_B} = R_X^{\mathsf{T}} \mathsf{H}^{AB} R_Y + \sum_{S\notin\{P,Q\}} \frac{\tilde{\mathsf{h}}^A_{XS}\tilde{\mathsf{h}}^B_{SY} + \tilde{\mathsf{h}}^B_{XS}\tilde{\mathsf{h}}^A_{SY}}{E_0 - E_S}. \tag{72}$$

  This is quasi-degenerate (Löwdin) perturbation theory. Its first term is term (i) at $W = \frac{1}{2}(R_X R_Y^{\mathsf{T}} + R_Y R_X^{\mathsf{T}})$, linear like everything else. Together with $\tilde{\mathsf{h}}^A_{XY}$ it is the second-order topography of the seam: what an intersection-space Hessian at a MECI needs (`geometry::optimize_meci`, §10, is the consumer), and what the branching plane leaves out.

Degenerate by symmetry. QD states in an E or T irrep are exactly degenerate, and term (ii) is $0/0$.

- **Jahn–Teller:** a linear splitting, $\tilde{\mathsf{h}}^A_{XY} \neq 0$. A cone, with no Hessian for either component.

- **Renner–Teller and the like:** no linear splitting. The components are smooth, and their Hessians are the eigenvalues of Eq. (72)'s matrix, not Eq. (60).

Either way Eq. (60) refuses a degenerate E_P and never divides: `df_qd_eigenvector_response` and `df_qd_sc_nevpt2` refuse a QD gap under `mcscf::kRootGapFloor`, naming both states.

12.3 The reference gaps: artefacts, in QD's own domain

Within the ensemble. At a crossing of two averaged CASSCF roots, the roots rotate by 90° over a width of about the gap. QD-NEVPT2 in the form the tree computes (Park's "SA-NEVPT2", with no extended-invariance rotation) is not invariant to that rotation, so E_P changes by $O(1)$ of the non-invariance over that width. It has a cusp at the reference intersection even where the QD gap is wide open. Eq. (64) is that cusp's $1/\text{gap}^2$ curvature, computed exactly. This bites where QD is used most: near-degenerate references are the reason to run QD at all, and the reference seam is generally not the QD seam. A Hessian there is the method's, not the molecule's. The tree refuses two roots under `mcscf::kRootGapFloor`, the gradient's #892 refusal, and a rotation $|\theta^A|$ above 10 per bohr (§11.2), and reports the smallest root gap and $\max|\theta^A|$; a threshold set by the cusp's own curvature has not been measured. The remedy is an invariant QD formulation (an XMCQDPT2/XMS-style rotation of the model space by f^{SA} before the dressing [7]). It would add one more closed-form block to x_A, and it is out of scope here.

At the ensemble's edge. The CI block of the SA-CASSCF Hessian has $2w_r(E_k - E_r)$ for every excluded root k, smallest at $r = n$, $k = n + 1$. As that gap closes, the solve degrades. Where it closes, the excluded root enters the average and the SA-CASSCF itself is discontinuous. Both routes fail there. `df_qd_sc_nevpt2` finds E_{n+1} by one `ci::solve` of $n + 1$ roots at the solution's orbitals, refuses an edge under `mcscf::kRootGapFloor` and reports it; the pipeline's numerical method logs it at every displacement.

12.4 Symmetric geometries

The setting: a symmetric reference geometry R_0, states M and N of the model space, and a coordinate q whose irrep lies in $\Gamma_M \otimes \Gamma_N$. Then q can couple M and N.

- **The totally symmetric block** is as well defined as at $n = 1$. Along a totally symmetric q no symmetry-empty perturber is populated, and states of different irreps do not couple.
- **Diagonal elements** carry the single-state per-root failures into term (i): SC's ρ_τ and PC's lift, at second order in the perturber.
- **Off-diagonal elements** fail one order earlier, in the coupling itself, when the model space holds irreps $\Gamma_M \neq \Gamma_N$ and some vibration carries $\Gamma_M \otimes \Gamma_N$. Both variants do, for the same reason. Mixing irreps is not enough: H_2CO's A_1/A_2 pair has no a_2 vibration to couple it.

SC. Take a tuple t empty for ket N by symmetry ($\Phi_t^N = 0$) but populated for bra M. SC's perturber is one function per tuple, so Cauchy–Schwarz bounds $|T_{MN}^t| \leq (N_t^M N_t^N)^{1/2}$. Along $h\,q$, $\Phi_t^N \approx h\,\partial_q \Phi_t^N$, so

$$T_{MN}^t = O(h), \qquad \frac{H_t^N}{N_t^N} \to \rho_t(q), \qquad C_{MN}^t \approx -\frac{h\,\langle \Phi_t^M | \partial_q \Phi_t^N \rangle}{\Delta_t + \rho_t(q)}, \tag{73}$$

with $\rho_t(q)$ the Rayleigh quotient of $\partial_q \Phi_t^N$ in the Dyall tables. This is odd and homogeneous of degree 1 in q, but not linear, unless ρ_t is constant over the coordinates of q's irrep. That holds on a single coordinate, for parallel derivatives, for Sijrs ($H = 0$), and for a one-function class (Eq. (5): $\rho_t = \eta$ whatever the direction). With the roles swapped (empty bra, populated ket) the denominator is the populated ket's and the term is linear. But H^{eff} averages C_{MN} and C_{NM}, so either state's empty tuples count.

- **The functional.** H^A_{MN} at R_0 is a directional derivative, not a gradient. Term (ii) along such coordinates is $2\tilde{\mathsf{h}}(q)^2/(E_P - E_Q)$, a degree-2 function that is not a quadratic form, so that block has no Hessian. The coupling $\mathbf{h}_{PQ}$, and the NAC $\mathbf{h}_{PQ}/(E_Q - E_P)$, are ill-defined there too.

- **The tree's computed coupling.** It drops a perturber by the *ket's* norm, $|N_t^N| \leq 10^{-14}$ (§14, SC's guard). At R_0 the reported $\mathbf{h}_{PQ}$ therefore omits this term without saying so, and it is added back once $|h\,\partial_q \Phi_t^N|^2$ passes the guard. `sc_effective_hamiltonian_tangent` still forms and reports a guarded tuple's $\dot{T}^t_{MN} = \langle \Phi_t^M | \partial \Phi_t^N \rangle$, which is what the refusal below reads.

- **The gradient is unaffected.** At R_0, $W^P_{MN} = 0$ across irreps, so dE_P/dR never reads H^A_{MN}, and nearby it reads it only at weight $O(h)$. The gradient stays C^1.

PC. The same thing happens through the lifted null modes of S_N: PC does not escape it. At R_0 an exact null vector v of S_N annihilates every row of a transition metric whose bra is N, so $v^{\mathsf{T}} b^{M \leftarrow N} = 0$ and C_{MN}'s *value* at R_0 is safe. Along $h\,q$, though, a spatial-symmetry null mode lifts. The function Gv (the internal functions combined by v) becomes $h\,w_v(q)$, with eigenvalue $\lambda = O(h^2)$. Normalise it, $\phi = w_v/\|w_v\|$. In an abelian group, symmetry allows $\langle \phi | V | M \rangle = O(1)$ and then forces $\langle \phi | V | N \rangle = O(h)$ (the two cannot both be $O(1)$ unless q is totally symmetric). The exact functional, with the lifted mode kept, therefore carries

$$C_{MN} \ni -\nu \, \frac{\langle \phi | V | M \rangle \, \langle \phi | V | N \rangle}{\Delta_t + \langle \phi | \hat{H}_D | \phi \rangle} = O(h) \times (\text{a function of } q\text{'s direction}), \tag{74}$$

which is SC's Eq. (73) with ϕ in place of the normalised $\partial_q \Phi_t^N$. On the diagonal the same mode gives $|\langle \phi | V | N \rangle|^2 = O(h^2)$: that is the single-state Schur lift. With the tree's absolute cut, the lifted mode stays dropped while λ is below the cut. The computed C_{MN} is then smooth, but it is missing an $O(h)$ piece, and E_P misses an $O(h^2)$ piece through term (ii), Hessian-sized and unreported. Once λ crosses the cut, the piece switches on with a jump.

One refusal covers both, on every root. The off-diagonal failure needs exactly what the diagonal one needs, SC's ρ_τ defect on the pair of coordinates or PC's Schur lift, and in addition a non-zero first-order overlap with some bra in the model space: SC's $\langle \Phi_t^M | \partial_q \Phi_t^N \rangle$, the tangent of T^t_{MN} at one dotted ket; PC's $\langle \phi | V | M \rangle$. So the existing refusals run on *every* model state's guarded tuples and metrics, whatever W^P weighs it: a state with $W^P_{NN} = 0$ still enters through $\mathbf{h}_{PN}$. A defect with a non-zero transition tangent refuses that (pair, coordinate pair) block; one with a zero tangent is the single-state term (i) case.

- **SC** (`sc_effective_hamiltonian_hvp`, #962): with partners, each ket's empty tuples that both directions populate are scored twice, by the single-state weighted defect over that ket's tables (`defect`) and, per bra M, by ρ_τ weighted by the tuple's share of coupling (M, N)'s first order, $\max(|\dot{T}_u|, |\dot{T}_v|)/|\Delta + \eta|$ against the largest over ket N's tuples (`coupling_defect`). Either past 10^{-6} (`kEmptyDefectTolerance`) refuses, Numerical, naming the coupling, the

class, the tuple and the directions. `df_qd_sc_nevpt2` takes every pair of coordinates that both populate an empty tuple of some model state as each other's partners.

- **PC** (`qd_pc_hessian_product`, #964): per direction and ket N, the Schur lift L of N's null modes along the root set's path ($\Gamma''(c_N, c_N) = \Gamma(Ku_N, c_N) + \Gamma(c_N, Ku_N) + 2\Gamma(u_N, u_N)$) and, per bra M, how those modes reach M's rows at first order (`overlaps`, in units of the null space's own round-off, over the modes that reach the cut within 10^{-3} along the direction, `kPcLiftStep`). A pair (M, N) whose overlap is past 100 (`kPcLiftOverlap`) along the direction and along its partner refuses that block, Numerical, naming the coupling, the directions, the class and both t^*. A lift with no bra overlap is the single-state one, reported by t^*.

A model space within one irrep, or one whose irreps no vibration couples, never meets the off-diagonal case by symmetry, only by accident: a perturber empty at an accidental zero of its amplitude. That is what #894's SC-QD miss at ethylene's Py was, a one-function perturber the guard dropped, and #949's one-function classes removed it (Eq. (5)).

Measured on the C_{2v} A_1/B_2 case. `h2nb_sto3g_sa2` (#1058) is H_2N–B, STO-3G, CAS(4,3) over the frontier π, σ and empty in-plane p, SA-2 over $1A_1$ and $1B_2$, whose two b_2 vibrations couple the roots. SC refuses nothing: every tuple empty for one root and populated for the other is in Sij, a one-function class, so Eq. (73)'s term is linear, and C_{01}'s ladder is bilinear on (B_2 1, B_2 2) to 1.7×10^{-10} of an allowed 8.2×10^{-9}. PC refuses exactly (B_2 1, B_2 1), (B_2 1, B_2 2) and (B_2 2, B_2 2), on the coupling $(0, 1)$ through $1B_2$'s Sijr, $t^* = 1.5 \times 10^{-5}$ bohr along each, as the fixture's analysis expects ($1B_2$'s Sijr, Srs and Sir modes lift as $\lambda/\sigma_{\max}/s^2 \approx 2$–$5 \times 10^{-4}$, with $\langle\phi|V|1A_1\rangle = O(1)$ and $\langle\phi|V|1B_2\rangle = O(s)$). Every other block, the totally symmetric one included, is accepted.

Degenerate inactive orbitals. Every non-abelian point group at its symmetric geometry has them (e and t orbitals: NH_3, CH_4, benzene, any linear molecule's π). A coordinate that splits the pair leaves κ^A_{oo} undetermined: the pseudocanonical orbitals of a degenerate block depend on the direction of the split. The tree's PC is invariant to that rotation only at exact degeneracy, and SC not in general. The single-state Hessians refuse this (§7.5: gap below $10^{-6}\,E_h$ with $|f^{(x)}_{pq}| > 10^{-8}$), and `df_qd_nevpt2_tangents` applies the same rule at f^{SA} (on C_{3v} ammonia's SA-2 CAS(8,7) in 6-31G, SC refuses and PC records 22 pairs). A semi-numerical Hessian there is suspect for the same reason: the step picks the orbitals. That is `pipeline/hessian`'s growing-z^c warning.

12.5 Summary

13 The QD Hessian: algorithm, checks and status

The assembly. `df_qd_sc_nevpt2` is `df_sc_nevpt2`'s assembly (§7.4) part for part, in the same parametrization and coordinates, over

$$\mathcal{L}[W] = E_\rho + (Z, z, \Theta)\cdot g_{\mathrm{SA}} + \sum_{r<s}\zeta_{sr}\langle c_s|H|c_r\rangle + \mathcal{E}^{\mathrm{corr}}[W] + \Phi^{\mathrm{SA}}, \qquad x_A = (\kappa_{\mathrm{nr}}, \kappa_{oo}, \xi_r, \theta_{sr}). \quad (75)$$

$E_\rho = \sum_r \rho_r E_r$ takes E_{CAS}'s place. It is not stationary in the SA parameters (the multipliers make $\mathcal{L}$ so), so its share is not `df_casscf`'s whole Hessian but its three places, as `df_sacasscf`'s root I at one-hot weights: E_{xy} at the ρ-weighted RDMs and $F^{[x]}$, the CP right-hand side at ρ against x_B in both orders, and $x_A^{\mathsf{T}}\tilde{H}_\rho x_B$ (`RelaxedHessian::apply_coupled` at ρ: CI rows $\times\, 2\rho_r$, rotation rows $\times\, 2(\rho_r - \rho_s)$). x's layout is `ThirdDerivative`'s with *every* pair's rotation, at all-equal weights

region	Hessian of E_P	analytic route
C_1, or abelian-symmetric totally symmetric block; every gap of Table 1 open	exists	tractable; the reference case
abelian-symmetric, model space in one irrep, full Hessian	exists unless the per-root refusals fire	tractable where they pass
abelian-symmetric, model space with irreps $\Gamma_M \neq \Gamma_N$ that a vibration of $\Gamma_M \otimes \Gamma_N$ couples, along those vibrations	may not exist (§12.4)	refuse per block on every root's ρ_τ / lift with a non-zero transition tangent
near a QD crossing	exists, large term (ii)	tractable, and better than finite differences; harmonic frequencies unreliable
at a QD crossing	none (cone)	pair mean Eq. (71) and seam 2×2 Eq. (72) instead (#967)
symmetry-degenerate QD states	none (JT) or by Eq. (72) (RT)	refuse Eq. (60)
near a reference-root crossing	exists, $1/\text{gap}^2$, an artefact	computable, not physical; refused at the root-gap floor and $\lvert\theta\rvert > 10/\text{bohr}$
ensemble edge, active-space instability	degrades, then discontinuous	refuse; both routes fail
degenerate inactive orbitals (non-abelian groups)	not determined	refuse, as the single-state Hessians do
higher roots, diffuse bases	exists	$1/d^3$ near-intruders; tractable only with the smallest $\lvert d\rvert$ reported
PC, large active space	exists	on-shell residual amplified by S^+; tight CI convergence

Table 2: Where an analytic QD-NEVPT2 Hessian is defined, and where it is worth computing.

Algorithm 6 Analytic DF-QD-NEVPT2 nuclear Hessian of state P (`df_qd_sc_nevpt2`; PC's, #966, the same over PC's walk)

Input: converged DF-SA-CASSCF, all-electron; P; coordinates

1: refuse if two CASSCF roots, $E_{n+1}-E_n$ or $\min_{Q\neq P}|E_Q-E_P|$ is under `kRootGapFloor` (Table 1); H^{eff}, R, $W^P = R_P R_P^{\mathsf{T}}$
2: x_A, $A = 1..3N_{\text{at}} \leftarrow$ `df_sacasscf_cp`, then Eq. (64), then κ_{oo} at f^{SA} (`df_qd_nevpt2_tangents`); t_A; refuse $|\theta| > 10$/bohr and a split degenerate inactive pair
3: $\lambda[W^P] \leftarrow$ the gradient's Lagrangian at W^P (`qd_sc_lagrangian`); refuse a coordinate mixing an inactive pair within 10^{-3} bohr
4: $\langle t_A, \text{HVP}\, t_B\rangle \leftarrow$ `qd_sc_hessian_product` at W^P along each t; coordinates populating an empty tuple again as each other's partners ▷ Eq. (66) level 1, z^c; §11.4's NEVPT2 parts
5: E_ρ's share: E_{xy} at ρ, the CP right-hand side at ρ against x_B both ways, $x_A^{\mathsf{T}}\tilde{H}_\rho x_B$
6: (Z, z, Θ)'s terms $\leftarrow$ `df_sacasscf_z` and `ThirdDerivative` over x; ζ's three shares $\leftarrow$ `df_qd_zeta`
7: skeleton $\leftarrow$ `df_adjoint_xy`; $\tau\kappa$, $\kappa\kappa$ and the κ_{oo} rows as at $n = 1$ ▷ Eq. (39)
8: term (i) $\leftarrow$ the sum (Eq. (61)); term (ii) $\leftarrow$ Eq. (60) from the $\mathbf{h}_{PQ}$ (`df_qd_eigenvector_response`'s, one shared solve)
9: on *every* model state: SC's ρ_τ and PC's Schur-lift tests; refuse a block where a defect has a non-zero transition tangent to a bra (§12.4)
10: report the asymmetry and translational sum, every gap of Table 1, the smallest $|\Delta_t + H_t/N_t|$, $\max|\theta^A|$, $\max|\kappa^A_{oo}|$, $\max|\zeta|$, $\max|z^c|$, term (ii)'s share, and every part

too: x_A moves an equal-weight rotation, and E_ρ, the level-2 term and ζ's term all see it. The κ_{oo} rows' derivation carries over: E_ρ and $\langle c_s|H|c_r\rangle$ are invariant under a core–core or virtual–virtual rotation and g_{SA} is covariant, so each leaves only BCH's $-\frac{1}{2}\nabla F\cdot[\kappa_{\text{nr}}, \kappa_{oo}]$, and by the Lagrangian's orbital stationarity the rows are $\frac{1}{2}\langle\kappa^B_{oo}, [X_N, \kappa^A_{\text{nr}}]\rangle + (A \leftrightarrow B)$ at $N = \mathcal{E}^{\text{corr}}[W] + \Phi^{\text{SA}}$, as at $n = 1$. A coordinate whose total tangent is below 10^{-6} of the largest (a rigid translation, whose t is the CP residual and round-off) breaks no symmetry. The result is exactly symmetric, with the asymmetry before symmetrising returned; the W overload returns term (i) alone.

The semi-numerical QD Hessian (#959). Under a `quasi_degenerate [nevpt2_gradient]`, `[hessian]`'s numerical method is Algorithm 3 over `qd_sc_gradient` or `qd_pc_gradient`'s state form, state P by `state` (H^{eff}'s eigenvalues ascending at the reference), over `[casscf]`'s `nroots` at its `weights`. The state is tracked, not ranked: at each displacement each root is signed against the reference's by CI overlap, which takes the displaced eigenvectors u_k into the reference's root basis, and `tracked_qd_state` picks the k of largest $|\langle R_P|u_k\rangle|$. A best overlap that misses 1 by 10^{-3} or more, or a second state within 10^{-3} of it, is refused, naming the geometry. Each displacement logs the tracked state, the QD gap, the smallest CASSCF root gap, the ensemble edge (a CASCI of $n+1$ roots over the solve's orbitals), $\max|\zeta|$ and the NEVPT2 chain's hazards; the worst of each is `hessian.json`'s `qd`. The analytic method refuses a `quasi_degenerate` chain (Unsupported) until #968. The references in `test/gtest/data/qd_nevpt2_hessian/` are this algorithm's over the same gradient (a fourth-order stencil at h, $h/2$, $h/4$; SC at 3.125×10^{-4} bohr and PC at 2.5×10^{-3} on the skew H_2CO, since SC reaches its asymptotic range only at the smaller step), written by `tools/nevpt2_hessian/qd_nevpt2_hessian.py` over the C++ driver `free-breakfast-qd-nevpt2-hessian`.

check	what it pins	status
$n = 1$: equals the single-state Hessian (`df_sc_nevpt2`, `df_pc_nevpt2`)	reduction: term (ii) empty, $W = 1$, no θ or ζ	by construction in the walks (each is the one-state walk at $n = 1$); not run on the whole Hessian
Eq. (69)'s three Euler zeros, rank-one (H, Δ), and its $M = N$ reduction to Eq. (37); Eq. (70) at $M = N$	the off-diagonal finishes	identities of the formulas, not gated; the forward tangents against the reverse passes by duality, 10^{-10} (`Nevpt2QdTangent`, `Nevpt2QdPcTangent`), and the HVPs against ladders of the bars, 10^{-7}, with the off-diagonal and PC's three kinds of term each visible (`Nevpt2QdHvp`, `Nevpt2QdHessianProduct`, `Nevpt2QdPcHessianProduct`)
sum rule Eq. (62)	term (ii)'s antisymmetry, linearity of term (i) in W	term (ii)'s half returned as a diagnostic, zero by construction (2×10^{-18}); the whole sum not run
term (i) at a general symmetric W against differences of $\mathcal{E}[W]$'s gradient at that fixed W	Eq. (61) without the eigenvector term	the pair sums $W^P + W^Q$ are in the references (#953); not yet gated against the W overload
term (ii) alone: Eq. (59) against differences of R_P, phase-aligned	the $\mathbf{h}_{PQ}$ inputs and the eigenvector derivative	the list call's $\mathbf{h}_{PQ}$ against each pair's own solve, 10^{-9} (`HessianDfQdEigenvectorResponse`); term (ii) through the whole Hessian; dR_P against differences not run
`h2co_ccpvdz_sa2` (C_{2v}, A_1/A_2): term (ii) = 0 in every block, since H_2CO has no a_2 vibration	the decoupled case: the model space mixes irreps, but no coordinate couples them	not run (the case is term (ii)'s degenerate-pair refusal)
a C_{2v} SA2 with A_1/B_2 roots: ladders of C_{MN} along two b_2 coordinates and their sum, and the per-root refusals	§12.4's off-diagonal failure, SC and PC; that the refusal fires where the ladder is not bilinear	`h2nb_sto3g_sa2` (#1058): SC's ladder bilinear, nothing refused by the product, the Hessian (`CoupledA1B2RefusesNothing`, $\mathbf{h}_{01}$ against the ladders) or the pipeline; PC's product refuses exactly the expected $B_2 \times B_2$ blocks. Whole-Hessian references, #1093
Eq. (71) at the ethylene MECI against differences of $E_0 + E_1$'s gradient	the pair-mean Hessian at a cone	references in the tree (#1059: SC and PC, pair sum, seam W, third state); the analytic side, #967
`h2co_skew_ccpvdz_sa3` (C_1), equal and unequal weights	θ in and out of the solve, Eq. (67)	SC: states 1 and 2 at equal weights and state 1 at 0.4/0.2/0.4, every element within 2× bar (worst 0.010–0.030 bars), every part and θ's $\theta\theta$ block above 10 bars (`HessianDfQdScNevpt2`); PC, #966
semi-numerical QD Hessian (Algorithm 3 over the QD gradient, roots tracked by R_P's overlap)	the whole thing	the references (#953, #1060; `tools/qd_nevpt2_hessian`) and the pipeline's numerical method against them (`PipelineQdNevpt2Hessian`)

Table 3: The QD Hessian's checks, and where each stands.

Status. In build order: (0) the semi-numerical QD Hessian, the oracle, with the bilinearity check of Eq. (42) because of §12.4 (#953, #959, #1058, #1059, #1060); (1) the off-diagonal finishes and the forward tangents (#954, #958, #960, #962, #964); (2) term (ii) from the couplings (#957); (3) the pair-walk level-1 HVP at general W (#963, #964); (4) term (i)'s assembly over `df_sacasscf_cp`'s responses and `df_sacasscf_z`'s terms at $\lambda[W]$, with the ζ Hessian (#955, #956, #961) and SC's whole Hessian (#965); (5) Eq. (60) gated against (0), SC on the C_1 cases. SC went first, since #949 had removed #894. Not yet landed: PC's Hessian (#966), the crossing's Eqs. (71) and (72) (#967), the A_1/B_2 whole-Hessian references (#1093), and the analytic `[hessian]`, its frequencies and example (#968).

Cost. One block CP-SA-CASSCF solve over $3N_{\mathrm{at}}$ right-hand sides, shared by every P. Per wanted P: $3N_{\mathrm{at}}$ pair-walk HVPs (each about $n(n+1)/2$ single-state ones, the gradient's own scaling) and $3N_{\mathrm{at}}$ forward-over-reverse tangents. The $n-1$ couplings are one shared solve. This is the same order as the semi-numerical route's $6N_{\mathrm{at}}$ QD gradients (eight per coordinate under the ladder). Measured: 25–26 s a call on the skew H_2CO (cc-pVDZ, CAS(4,3), three roots, 12 coordinates) on the gfx1200.

14 Numerical hazards and refusals

- **SC's inactive non-invariance.** SC is not invariant to rotations among inactive orbitals, so z^c divides by orbital-energy gaps (Eq. (24)); a degenerate pair with a finite source is refused. PC's source vanishes with the gap, and its z^c keeps a finite limit there. Park finds SC's gradients unstable for this reason and recommends PC [8].

- **SC's small-norm pole.** A perturber whose own norm is tiny (around 10^{-8} on some symmetric cases) has a Rayleigh quotient H/N that round-off moves, and $c_t = -1/(\Delta + H/N)$ amplifies it by $\sim 1/N$. The energy cannot see it (cN is small); the bars and the PT2 density can. PC divides by no norm and is unaffected. The one-function classes no longer have it: their η (Eq. (5)) reads no N_t.

- **SC's guard.** A perturber under $|N| = 10^{-14}$ is dropped, so SC's energy kinks where a norm passes through zero and is not smooth at a symmetric geometry. The gradient is the one-sided limit; the Hessian refuses the pairs on which it is no bilinear form (§7.5). For the one-function classes, where the guard dropped a perturber whose QD couplings are linear in its amplitude, it is gone (#949).

- **PC's metric cut.** A real mode crossing the cut changes the rank, and the energy by a step of that mode's size (under $10^{-9}\,E_h$ where measured); a non-positive $\lambda + \Delta$ is refused. Near either, the energy is not smooth, and the derivatives inherit the solve's conditioning. First derivatives tolerate kept modes just above the cut; second derivatives do not, hence the floor (§4.3's "conditioning floor") and the Hessian's refusals of a thin kept spectrum and a nearby Schur lift.

- **Inactive near-degeneracies.** z^c and κ_{oo} divide by inactive orbital-energy gaps. In the gradient that is SC's instability and PC's finite limit; in the Hessian κ_{oo} enters quadratically. SC refuses a coordinate whose avoided-crossing half-width is under 10^{-3} bohr. PC's $1/\mathrm{gap}^2$ terms cancel, at a cost in digits, down to the $10^{-6}\,E_h$ degenerate-pair rule.

- **Gaps.** QD states closer than $10^{-8}\,E_h$ are refused by the state form, the coupling and the NAC (not by the W forms or the intersection); CASSCF roots closer than that which ζ, t

or T^{PQ} couple are refused, except PC's intersection inside the QD pair's own CASSCF pair, which returns the mean gradient alone (§10).

- **The QD Hessian's divisions.** Each gap the gradient divides by, the Hessian divides by once more (Table 1), and the QD gap enters for the first time, through term (ii). The QD gap, the CASSCF root gaps and the ensemble's edge are refused under `mcscf::kRootGapFloor`, a root rotation $|\theta^A|$ above 10 per bohr, and, on every model state, a block where a symmetry defect has a first-order overlap with some bra (§12.4); the rest are reported.
- **Reproducibility.** The einsums and the DF derivative kernels sum with atomics, so two runs differ at the 10^{-12} relative level (more where a solve is ill-conditioned); `rdms` and its adjoint are bitwise reproducible.
- **Convergence.** Every derivative assumes a converged SA-CASSCF; the gate measures how far the Lagrangian is from stationary, and the gap sensitivities bound the effect on a QD gap.

15 Verification

Each acceptance bar is fixed by analysis and checked once: 10^{-10} for two paths to the same number, 10^{-7} relative for a Richardson ladder of central differences (successive changes falling by a factor in $(3.2, 4.8)$ over three halvings), and twice the committed error bar for a numerical reference.

quantity	checked against
SC energies	PySCF DF-NEVPT2, every class, $10^{-7}\,E_h$
PC energies	block2 `WickICNEVPT2`, every class and dropped count
SC-, PC-QD H^{eff}	a brute-force reference: every element's per-class coupling, 10^{-10}
dm3, f3ca, f3ac	PySCF, 10^{-12} of max; tiled vs one tile, 10^{-13}
RDM adjoints	polarisation identities (the build is quadratic in c, linear in $(tu\|vw)$)
class reverse passes	Eq. (16), homogeneity and Euler's degree 4, 10^{-12}; ladders along B, h_v, ε, h, RDMs
$(\bar{h}, \Lambda, y, A)$	ladders along h, B, each rotation block and c, every input rebuilt
z^c	stationarity of $E_{\text{corr}} + \Phi$ along core–core and virtual–virtual rotations
Lagrangians	stationarity ladders (orbitals, CI off every root, rotations among roots) at equal, unequal and mixed weights; they pin $1/(2w_r)$
gradients	numerical gradients, 2× bar; translation and rotation sums
W form	$G(W_1 + W_2) = G(W_1) + G(W_2)$, 10^{-10}
$\mathbf{h}_{PQ}$, $\mathbf{d}^{\text{NEV}} + \mathbf{d}^{\text{CAS}}$	numerical references, 2× bar; $\mathbf{h}_{QP} = \mathbf{h}_{PQ}$, $\mathbf{d}_{QP} = -\mathbf{d}_{PQ}$
$\mathbf{d}_{PQ}$, $\mathbf{d}^{\text{PT2}}$, D^{PT2}	a brute-force reference, 2× bar; 10^{-10} of max $\|D\|$
fold	the three-solve path, 10^{-9} (two independently converged solves)
intersection	each W form at the same u's; the gap bound on a stopped SA-CASSCF; inside a degenerate CASSCF pair, $\bar{\mathbf{g}}$ against directional differences of $\bar{E}$ at the exact-C_{3v} point
one-function η	$\|X - \eta Y\|/\|X\| \leq 6 \times 10^{-15}$ on a state's RDMs; the SC-QD checks that missed at ethylene's Py
t_x, `builders_tangent`	ladders along each input; f against `zc_adjoint`'s Φ
HVPs	ladders of the adjoint along t_B; Euler zeros of Eq. (37); PC at $f = 10^{-10}$ on PSB3; the Schur-lift report against a direct probe of the kept σ
SC, PC Hessians	semi-numerical Hessians of #803's fixtures (two for SC, four for PC), every element within 2× bar (worst 0.085–0.118 bars); each part above 10 bars; CP residual $\leq 10^{-8}$, asymmetry $\leq 10^{-9}$, translation sum $\leq 10^{-8}$
Hessian refusals	each reached by a real input: `nh3_cs_sto3g`'s $A'' \times A''$ block and `nevpt2.toml`'s water (SC); a thin kept class, HCN CAS(6,6)'s lift, C_{3v} ammonia's degenerate pair (PC, #943)
SA response, t_x	CP residuals $\leq 10^{-9}$; Eq. (64)'s condition on every pair and f^{SA}'s off-diagonal change, zero to 10^{-10} of the largest summand, at equal and partly equal weights
QD forward tangents	duality against `qd_sc_adjoint`, `qd_pc_adjoint` at a general symmetric W, 10^{-10} of the largest summand (skew's conditioning floor where larger); Eq. (65), 10^{-12}
QD HVPs	ladders of the QD adjoints' bars, every field, 10^{-7}; the curvature, the off-diagonal finish and PC's three kinds of term each above ten bars; K6 against $\mathcal{E}[W]$'s second difference
QD refusals	a constructed one-sided Srsi tuple (SC); `h2nb_sto3g_sa2`'s expected blocks, exactly (none for SC, $B_2 \times B_2$ for PC)
z, ζ terms at QD multipliers	Richardson ladders of $Z \cdot g_{\text{SA}} + \sum z_r \cdot g_{\text{ci},r}$ and of $\langle c_s\|H\|c_r\rangle$ itself, 10^{-7}; the root-set curvature's and Θ's shares, and ζ's every piece, above ten bars
term (ii)	the list call's $\mathbf{h}_{PQ}$ against each pair's own solve, 10^{-9}; a degenerate QD pair refused
SC-QD Hessian	the semi-numerical references of #953 and #1060 (skew H_2CO, states 1 and 2 at equal weights, state 1 at 0.4/0.2/0.4), every element within 2× bar (worst 0.010–0.030 bars), every part above 10 bars; nothing across irreps on `h2nb_sto3g_sa2`
semi-numerical QD Hessian	the pipeline's against the references, every element within 2× bar; SC's pairs on the A_1/B_2 case all bilinear

A converged residual says a solve answered the equations it was given, not that they were the right

ones, so each piece also has a check that reads no residual: forward/reverse duality ($\langle u, y\rangle = \dot{E}(u)$, $\langle \kappa, A\rangle = \dot{E}(\kappa)$, and along a seeded integral perturbation $\langle(\bar{h}, \Lambda), (\dot{h}, \dot{B})\rangle = \dot{E}$); the gate; $E_{\text{corr}} \to 0$, where every multiplier is zero and the gradient is the DF-CASSCF one; stationarity ladders of $\mathcal{L}$; and the translational and rotational sums.

The oracle. No program the tree can import has a NEVPT2 gradient, so the end-to-end oracle is the numerical gradient of re-converged energies, at the same settings and the same auxiliary basis (an approximation is tested against the same approximation). PySCF's DF-NEVPT2 gives SC, and block2's or this tree's energy gives PC. Each displaced CASSCF passes a continuity guard before it is differenced: its energy against the first-order prediction, its active subspace and CI vector against the reference's, and convergence. A displaced optimisation that lands on another stationary point gives a number from another reference. Central differences plus one Richardson step are good to about $10^{-6}\, E_h/a_0$, and that is the gate a consumer holds an analytic gradient to, not the tree's usual 10^{-7}, because the oracle is itself a difference. Near-degenerate inactive orbitals (SC's hazard, §14) are where it fails first: the analytic gradient is right there, but the energy curves too sharply for the differences.

16 Map to the code and to the papers

here	code
§4.1	`nevpt2::mo_blocks`, `h2e`, `slab`, and `*_adjoint`
§4.2	`nevpt2::rdms`, `rdms_adjoint` (one vector; bra/ket)
Eqs. (4), (6)	`class_energies`, `pc_class_energies`
Eqs. (7)–(8)	`sc_effective_hamiltonian`, `pc_effective_hamiltonian`
§5	`class_energies_adjoint`, `pc_class_energies_adjoint`, `*_effective_hamiltonian_adjoint`
Eqs. (30), (31), (33)	`response::zvector`; `mcscf::df_gradient`'s multipliers and extra adjoint
§6	`sc_energy`, `sc_adjoint`, `pc_*`, `zc_multipliers`, `zc_adjoint`, `lagrangian_gradient`, `sc_gradient`, `pc_gradient`
§8	`qd_*_energy`, `qd_*_adjoint`, `qd_sc_lagrangian`, `qd_*_gradient` (state, W, several W)
§9	`qd_*_coupling`, `*_pt2_transition_density`, `qd_*_nac`
§10	`qd_*_intersection`
Eq. (5)	`single_function_etas`, `single_function_eta_bars`
§4.3, the floor	`EnergyInputs::pc_floor`, `PcClassEnergies::cut`, `kept`
§4.5	`Nevpt2Step::run` (`pipeline/nevpt2`); `sc_couplings`, `pc_couplings`, `pc_kets`, `solve_class`
Eq. (36), Eq. (35)	`nevpt2::builders_tangent`, `hessian::df_nevpt2_tangents`
§7.3	`class_energies_hvp`, `pc_class_energies_hvp`, `sc_hessian_product`, `pc_hessian_product`
§7.4	`hessian::df_sc_nevpt2`, `df_pc_nevpt2`, `df_adjoint_xy`, `df_casscf`, `df_casscf_z`
§7.6	`pipeline/hessian`'s numerical method, ladder and bilinearity check; `tools/nevpt2_hessian`
Eqs. (59), (60) term (ii)	`hessian::df_qd_eigenvector_response`
§11.2	`hessian::df_sacasscf_cp`, `df_qd_nevpt2_tangents`; `nevpt2::qd_builders_tangent`
Eqs. (68), (69)	`sc_effective_hamiltonian_tangent`, `sc_effective_hamiltonian_hvp`
Eq. (70)	`pc_effective_hamiltonian_tangent`, `pc_effective_hamiltonian_hvp`
Eqs. (66), (67)	`qd_sc_hessian_product`, `qd_pc_hessian_product`; `hessian::df_sacasscf_z` at a QD Lagrangian, `df_qd_zeta`
§12.4	the products' `coupling_defect` (SC) and `overlaps` (PC)
Eq. (61), Algorithm 6	`hessian::df_qd_sc_nevpt2` (state and W forms); PC's, #966
Table 3's semi-numerical row	`pipeline/hessian`'s QD chain, `tracked_qd_state`; `tools/qd_nevpt2_hessian`, `tools/nevpt2_hessian/qd_nevpt2_hessian.py`

here	Park
Eq. (15), Eq. (16)	[8], Eqs. 36–38, 40
Eq. (24)	[8], Eqs. 56–57; [10], Eqs. 25–27
Eq. (29)	[8], Eq. 23
Eq. (9), Eq. (43)	[10], Eqs. 16–20 ($R_{PPMN} = W_{MN}$)
the QD reverse passes	[10], Eqs. 21–23 (λ, implicit in the bars), 31–43
Eq. (44)	[10], Eq. 24, with ζ and Θ explicit
the Z-vector right-hand sides	[10], Eqs. 28–29
§9	[10], Eqs. 45–54

Where this departs from Park.

- Park's QD gradient is PC only, with SC called straightforward [10]; both are implemented here, with SC's hazards stated.
- Park reports gradients and optimised geometries, not Hessians. The single-state Hessian of §7 and the QD Hessian of §11 have no counterpart in his papers; their only reference is the semi-numerical Hessian over the analytic gradient.
- SC's one-function classes keep every perturber at η (Eq. (5)) instead of PySCF's guard, so this SC differs from PySCF's and Park's where a one-function perturber's norm passes through zero (by at most $6.5 \times 10^{-13}\,E_h$ where measured).
- Park contracts 4-RDM derivatives through multipass intermediates ([10], Eqs. 55–58). Here the f3ca/f3ac build is differentiated directly, tile by tile, so neither the 4-RDM nor its derivative exists. Kollmar et al.'s factorisation of the Dyall matrix elements [12] would remove f3 altogether; it is not used.
- ε is a function of γ (or γ^{SA}) inside the functional, the multipliers among the roots (ζ, Θ) are explicit, and the reference block is the diagonal form $E_M \delta_{MN}$, equal to Park's on the constraint surface.
- Printed misprints affect a reader of [10]: Eq. 5 drops the two-electron integral of the active term; Eq. 16 omits $\langle M|H|N\rangle$; Eq. 29's $z^c_{xy}\langle I|E_{xy}|N\rangle$ is zero as printed and should be $[\mathbf{g}(z^c)]_{xy}$ ($\mathcal{W}$ here); Eq. 41 has $\Gamma^{(3)}_{a'a,b'b,ec}$ where Eq. 39 has $\Gamma^{(3)}_{a'd,b'b,ec}$; Eq. 56 sums over its free indices.

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